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Counting Little Fish For Science

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ABSTRACT

The delta smelt are an endangered indicator species that exists within the San Francisco Estuary. They are an extremely small and fragile species that range from 60-70mm in length [1]. Environmentalists are reliant on indicator species such as these to understand how the surrounding ecosystem is affected by pollution, invasive species and other environmental hazards. Due to their species status, understanding the true number of delta smelt in the wild is important for conservation efforts. However, their near translucent body makes tracking the fish extremely challenging for researchers especially as, even holding them briefly, can cause them to die [2]. The Wildlife, Fish, and Conservation Biology Department of UC Davis (WFCB) have been attempting to reintegrate delta smelt into the wild, but are still utilizing an inefficient way of counting the number of fish that they are transporting. We have paired up with the WFCB in order to design a new tracking system that will reduce the time it takes for them to count the delta smelt, and also reduce the likelihood of the fish dying to stress.

Our team decided on the usage of machine learning to tackle this issue. Currently, the research lab transfers the fish with a pump from the pond to an RV to then deliver into barrels for further transportation. After discussions with the research lab using ideas of previously existing products, measurements of already existing equipment, and calculations, we came to a final system design. Using the pump that the research team already uses to transfer the fish from the pond to their RV, we created a transfer unit that allows the fish to pass through before the RV container where a camera (GoPro Hero 7 Black) sends footage to a computer that analyzes the number of fish passing through using a python code based on YOLO. For stability and logistical reasons, the transfer unit was placed over the RV, flowing the fish into it using a hose. The report will focus on our design requirements, verifications, system architecture, failure analysis and engineering processes taken over the past two quarters.

SECTION 1: INTRODUCTION

Project Overview

Currently, the team utilizes a specialized fish pump to move the fish from their pond into a separate container where they are counted by hand. Once that lengthy process is completed, the fish are moved from the counting tank to the final barrel for transport back to their natural habitat. The team wants to automate the counting process. We decided to completely eliminate the fish being placed into the counting container directly to the final container, with the counting device being somewhere in between to count the fish. This reduces the amount of stress that the fish face while being counted. This also reduces the cost of the process as the pump would not be powered on for as long, reducing the need for fuel to power the generator. One possible solution we originally saw on the market was to use infrared light and cameras to track fish with high accuracy [3]. While this is a step in the right direction for a possible solution, it falls short as the method cannot count the fish, only works in water with low turbidity, and uses a very expensive camera outside of the budget provided to us.

A successful solution would aim to count the fish within 5% accuracy without harming the fish and being as non-intrusive to the fish as possible. The fish are extremely fragile which when described to us by the sponsors, would die if they hit the container 3 times. We must minimize the amount of times that the fish is in contact with any external surfaces or any additional design structures. They also mentioned, of 3000 delta smelt that they counted, about 300 died. Therefore, one of our design goals was to minimize death throughout the counting process, with a goal to at least maintain over 2700. Our design must also be portable as the team is often in the field. It must also have a display with numbers indicating the amount of fish that have passed through. To counter these problems, our team decided to use a Go Pro Hero Black 7 as it allows for video analysis as a stream rather than a recording. Moreover, the system utilizes a "grooved" lane system that separates the fish to facilitate tracking and with material that would be fish safe. Overall reducing the time it takes the team to count the fish, the amount of fish mortality and portable for the team to take to the field.

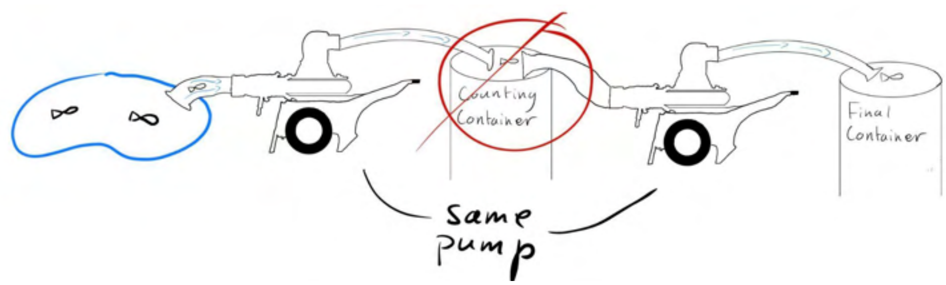


Figure 1: Sponsor System Design and Goal

Product Objectives

Based on the goals and problems that the WFCB discussed with us, our product objectives are as follows:

1) The system may be semi-autonomous
2) The budget is ~$1500
3) Our system must have a margin of error of 5% or less in terms of accuracy for fish tracking
4) 10% or lower mortality rate
5) Does not hold the fish in an additional container from the existing system
6) Must be portable via the means of the research lab

Stake Holder Analysis:

The two main stakeholders are the manufacturing side as well as the conservation field.

Conservationist

Our sponsor in this project is the Wildlife, Fish, and Conservation Biology Department of UC Davis. Through them, we understand a great deal about the declining population of the delta smelt and how valuable they are to the environment as an indicator species. With less time being spent counting the delta smelt, more time can be focused on the conservation efforts of groups like the WFCAB. The environment will also be easier to maintain with an indicator species helping alerting groups on invasive species and other harmful effects in the water.

Manufacturing
The stakeholders will work with the device itself in the manufacturing side. Creating a final version of our prototype would require many engineers and machinists. The engineers could spend time redesigning the prototype to maximize efficiency while the machinists could better reduce tolerances. Moreover, on the manufacturing side would also include any companies that were utilized to purchase the material used such as the PVC or the PLA filament.

Concept of Operations

The concept we utilized heavily relies on machine learning. The research team will pump the fish from a pond and then transport them into a tank where they can typically move the fish before filling them up in multiple canisters. Due to one of the requirements being to get rid of the intermediary step of holding the fish in another location, we designed our system to track the fish as they are being transferred from the pond to the tank. The fish will be continuously flowing rather than being held up in an additional tank where they could potentially get stressed and die, which is important for our mortality rate requirement. The pump flow rate is about 1 m/s so we will use different sized pipes to slow them down before the camera captures them.

The fish tracking system is separated into the subsystems: sensor, transfer unit, computation and the attachment. The sensor sends information to the computation system which calculates the number of fish that pass through the transfer unit. The camera lens, which is placed inside the transfer unit facing the fish, will capture them coming through the pump before entering the tank. The transfer unit is placed on top of the attachment which can be placed over the tank that the research team uses to transport the fish. The process can be seen in Figure 2.

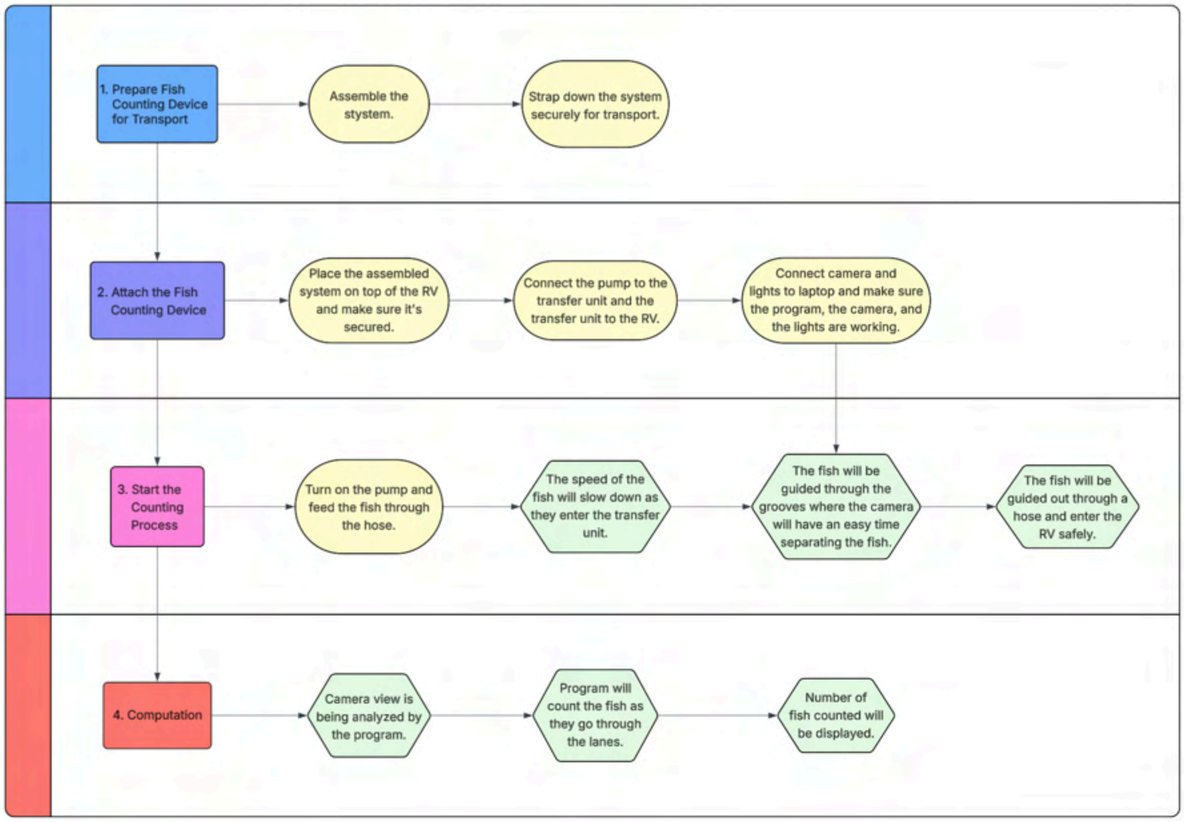


Figure 2: System Process Flowchart

SECTION 2: PRODUCT REQUIREMENTS & REQUIREMENT ROLLDOWN CHART

General Requirement:
GR. 1.0: The system may be semi-autonomous
GR. 1.1: The budget is ~$1500
GR. 1.2: Our system must have a margin of error of 5% or less in terms of accuracy for fish tracking
GR. 1.3: 10% or lower mortality rate
GR. 1.4: Does not hold the fish in an additional container from the existing system

Sensor Requirement:
SR. 1.0: Steadily Powered
SR. 1.1: Sensor must have high shutter speed
SR. 1.2: Sensor must have a minimum of 42 FPS to 104 FPS
SR. 1.3: Must be stable within transfer unit so that no shaking occurs

Computation Requirement:
CR. 1.0: Must be able to be used on a laptop for field work
CR. 1.1: Must be able to analyze the sensor footage

CR. 1.2: Must be able to differentiate between each fish

Transfer Unit Requirement:
TUR. 1.0: Under 130lbs
TUR. 1.1: Utilizes grooves to spread out the fish
TUR. 1.2: Must slow water down to 0.36 m/s or less
TUR. 1.3: Must hold sensor within Transfer Unit
TUR. 1.4: Must be stable on top of the attachment subsystem so that no shaking occurs
TUR. 1.5: Enough lighting to ensure visibility for the sensor

Attachment Requirement:
AR. 1.0: The attachment must hold the weight of the transfer unit (including water flow)
AR. 1.1: Must be portable
AR. 1.2: Must be stable to withstand water flow
AR. 1.3: Must be stable over any terrain so no shaking occurs

Roll Down on Requirements

Cost
The total budget given to us by our sponsors is $1500 (GR 1.1). As shown in Table 1, the total cost of everything is $1461.08. This does not account for the sensor which is a GoPro camera, which was donated to us for this project by our sponsors. This donation took a significant amount off the total cost and it is projected for us to spend about $1067.79, which is under the budget. The main cost of this system is building the transfer unit. This is due to the large size of the transfer unit, which uses a lot of material to build, and a way to reduce cost is to design the system in a smaller footprint.

Subsystem	Cost
Sensor (Donated)	$359.99
Computation	$0
Transfer Unit	$984.56
Attachment	$129.32
Shipping & taxes	~$103.03 - $136.33
Total Cost Including Donated Item Cost	$1610.20
Total Cost	$1216.91

Table 1: Cost Breakdown

SECTION 3: FINAL DESIGN SUMMARY

Complete System Overview

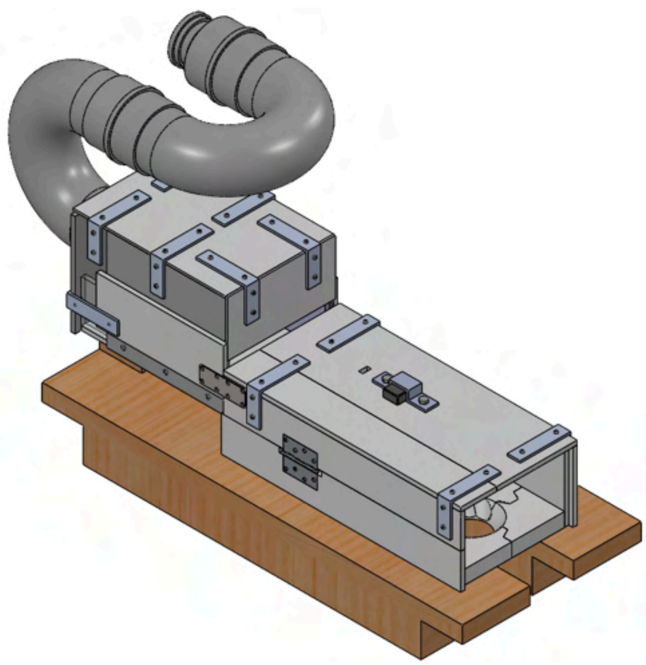


Figure 3: Isometric View of Fish Tracking System

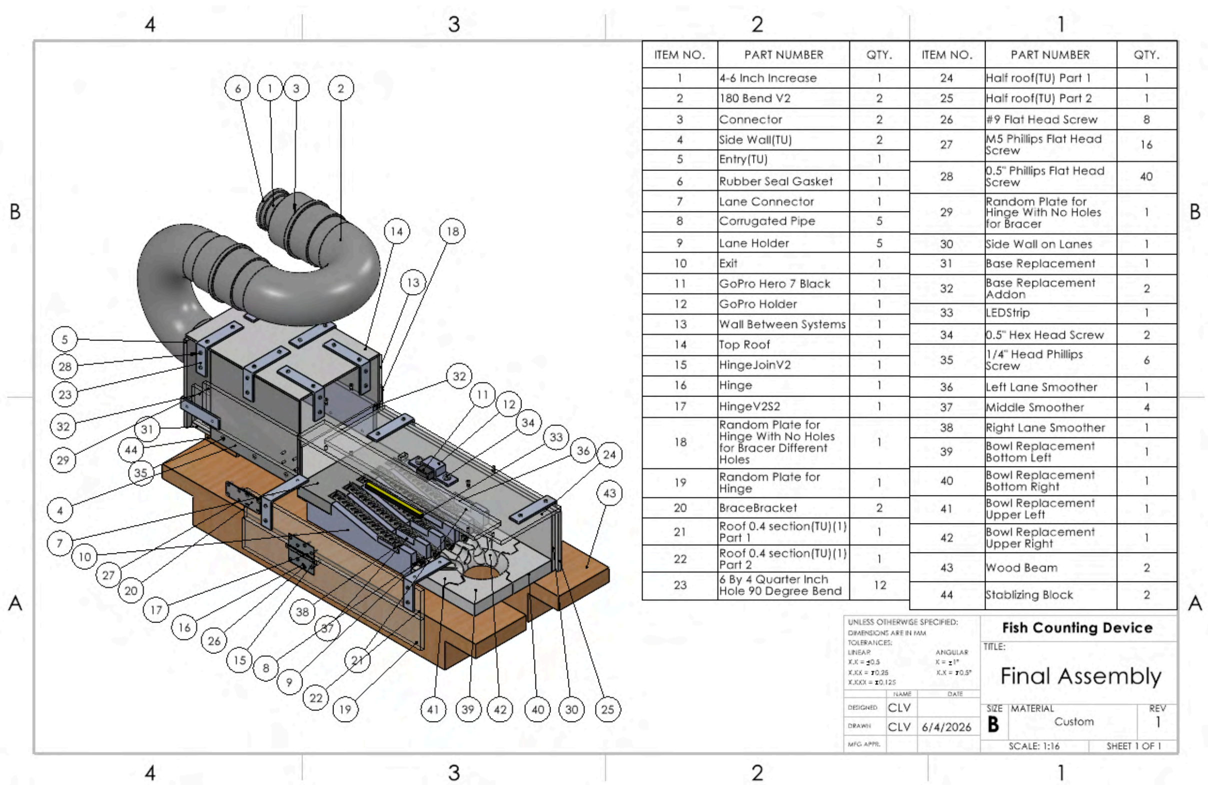


Figure 4: Drawing of Fish Tracking

The overall system consists of four subsystems namely the transfer unit (1-10,14-42), the sensor (11,12,34), the computation which is done using an external laptop, and the attachment (43,44), which holds the transfer unit above the tank. The sensor (GoPro) sends a video to the computer through which the code computes the number of fish passing through the transfer unit. The attachment design has changed from the initial design due to placement obligations not met by the initial design. Instead of holders, we incorporated long wooden planks that go across the length of the tank to provide support throughout instead of just the edges.

<u>Subassembly Overview</u>

Sensor Overview

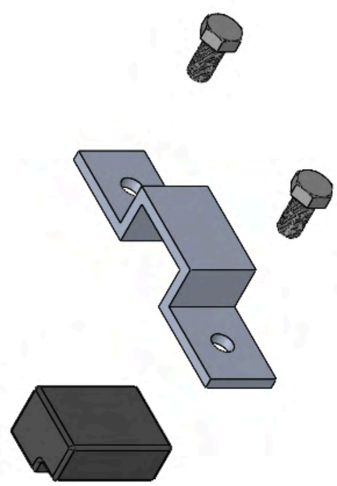


Figure 5: Exploded View of Sensor Sub-Assembly

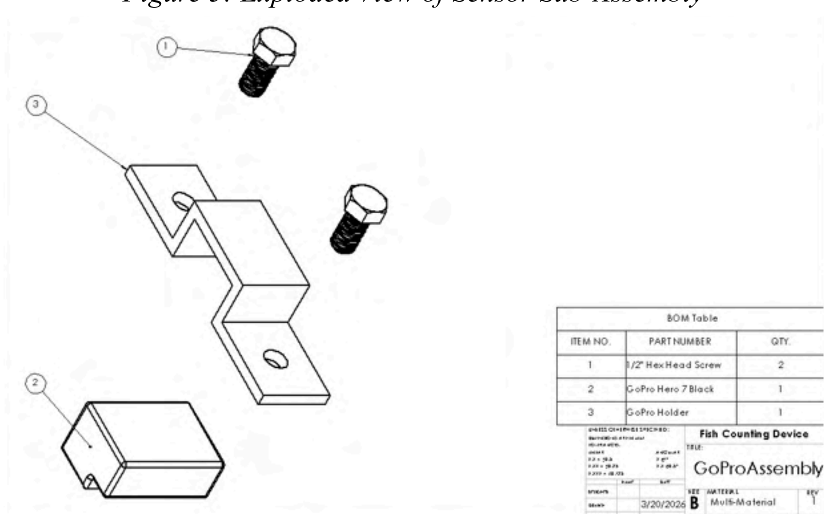


Figure 6: Drawing of Sensor Sub-Assembly

For the sensor portion of our design, we have selected to use a camera, the GoPro Hero 7 Black. This GoPro is lightweight, weighing about 0.26 lbs [3]. As the sensor will be attached onto the transfer unit, the lightweight feature of the GoPro helps meet the total weight requirement of the transfer unit (TUR. 1.0) being below 130 lbs. The computation system requires good footage and visuals of the fish passing through. Due to the speed at which the fish are coming through, we determined that the sensor must have between 42-104 FPS as per our calculations (Calculation 2) in order to meet our accuracy requirements of 5% (GR. 1.2). The GoPro Hero 7 Black can go up to 240 FPS depending on the quality of resolution, with a resolution of 1440p, the camera can go up to 120 FPS [4]. According to manufacturers, the GoPro has auto and manual shutter speed allowing us to adjust it to a higher speed manually if needed as seen in Figure 7. This helps with meeting our shutter speed requirement (SR. 1.1).

We utilize red lights in order to have the transfer unit well lit while the fish are passing through as it does not harm the fish [2] and thus meeting the requirement of having enough lighting (TUR. 1.5). There is a hole on top of the transfer unit enough for the camera portion of the GoPro to slot in and prevent shaking from having the GoPro hang or not have enough weight support. The camera will also have a fixture that can be screwed on top of the screen (without messing with the user interface) for additional support and increasing stability from the sides (SR. 1.4). Moreover, the GoPro is already waterproof, so any slight splashes of water should not affect the camera's functionality, however, the touch screen will not work with water hence our design only requires the camera lens to be facing water [5]. As we are able to utilize a camera that the research team already owned and allowed us to use, it will save ~$360 from our budget requirement (GR. 1.1).

For Video, the options depend on the fps setting, as shown below.

Shutter	Example 1: 1080p30	Example 2: 1080p60
Auto	Auto	Auto
1/fps	1/30 sec	1/60 sec
1/(2xfps)	1/60 sec	1/120 sec
1/(4xfps)	1/120 sec	1/240 sec
1/(8xfps)	1/480 sec	1/960 sec

Figure 7: GoPro Hero Black 7 Spec Sheet FPS Options [5]

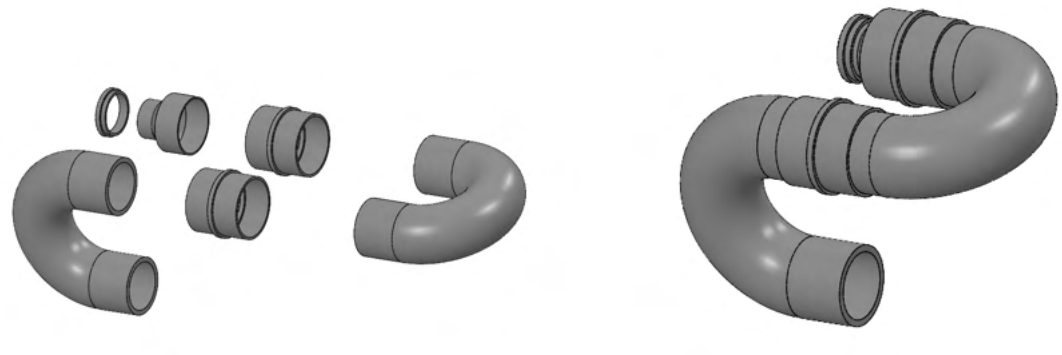
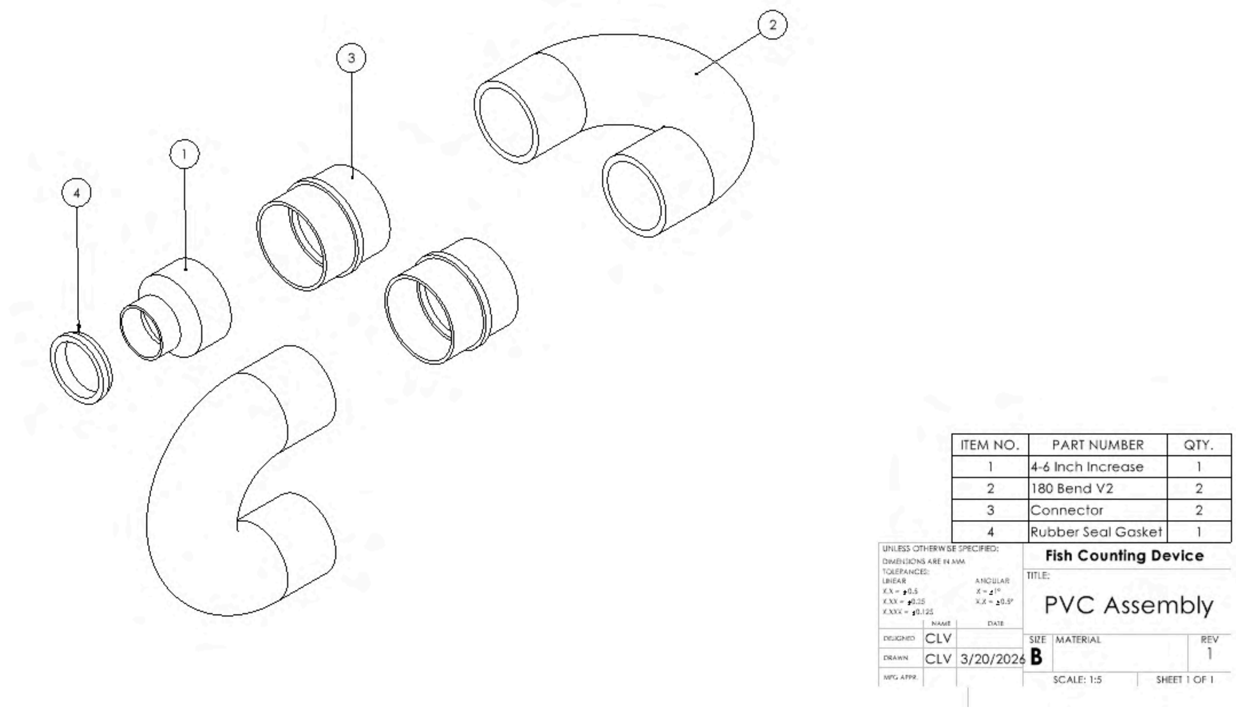


Figure 8: *Exploded View of PVC Sub-Assembly*



ITEM NO.	PART NUMBER	QTY.
1	4-6 Inch Increase	1
2	180 Bend V2	2
3	Connector	2
4	Rubber Seal Gasket	1

Figure 9: Drawing of PVC Sub-Assembly

Computation Overview

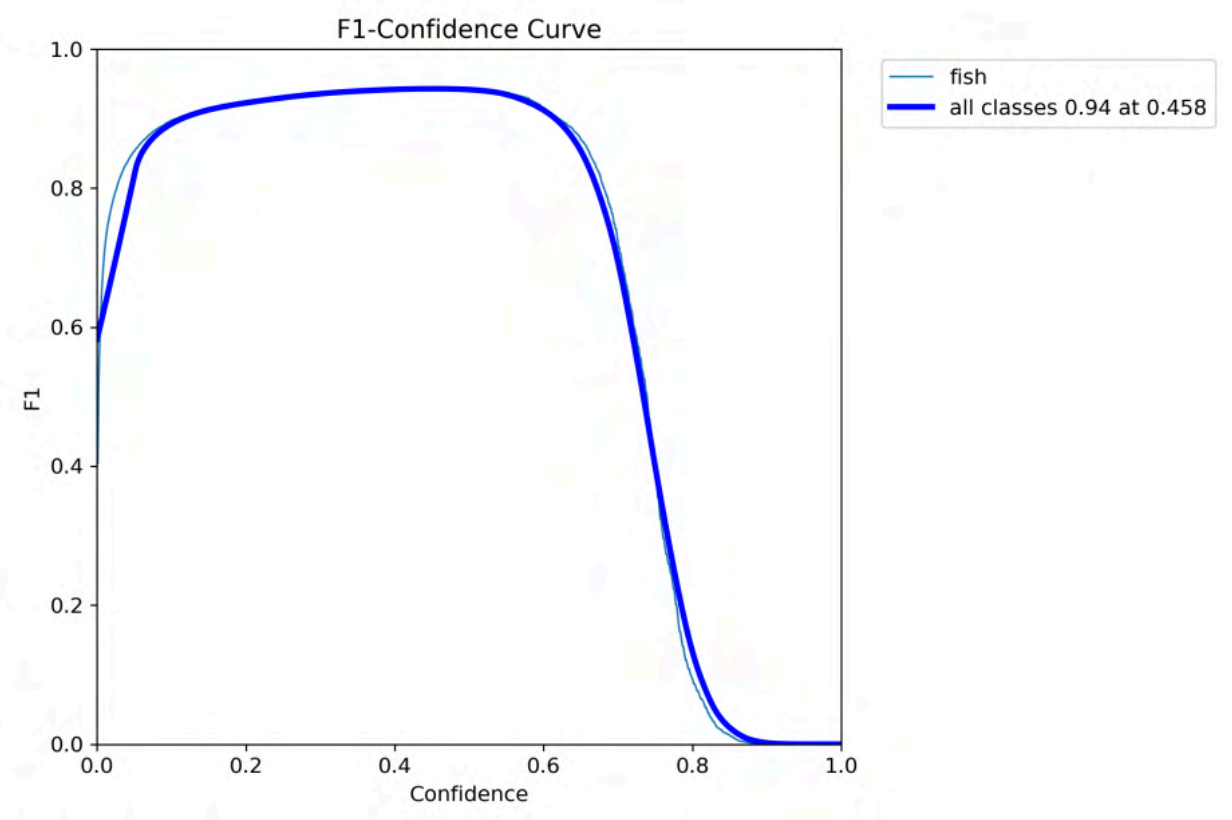


Figure 10: F1-Confidence Box Curve

The fish counting system used in this project is YOLOv8, a real-time object detection algorithm developed by Ultralytics. YOLO (You Only Look Once) models are widely used in computer vision because they perform object detection in a single direction through a neural network, enabling high speed detection suitable for real time applications. [6]

To improve training efficiency on the detection capability, the model was pretrained based on public online datasets like Kaggle. Pretraining allows the neural network to learn general features from fish before task specific training. Which shortened the time and training process required on limited customized data.

After the initial pretraining, fine tune was performed using manually labeled video frames extracted from fish transfer recordings. Bounding boxes were annotated around each fish using Python code, creating a custom dataset for this project. These annotations were used to fine tune the YOLOv8 model for accurate fish detection and counting during operation (CR. 1.1).

YOLOv8 was introduced in 2023 and includes an anchor free detection head, improved feature extraction architecture, and simplified training workflows. It improved detection accuracy while maintaining fast inference speed compared to earlier models such as YOLOv5 and YOLOv7, which is critical for applications requiring real time video processing. [7][8]

Despite the newer developments such as YOLO26, which introduces more architectural improvements on edge computing hardware. YOLOv8 was selected for this project because it provides a stable performance in the working environment in the current task. YOLOv8 offers several advantages that make it suitable for the fish counting system including: real time inference speed, high accuracy for detecting small moving objects, and moderate computational requirements (CR. 1.0). These characteristics are essential because the system must process approximately 1000 fish per minute while maintaining accurate counting performance (CR. 1.2). [9]

Alternative computer vision models were also considered during system design. For example, the Segment Anything Model developed by Meta AI, can provide pixel level segmentation of objects in images. While segmentation models produce highly detailed object boundaries, they require significantly more advanced computing hardware and processing time compared to object detection models. [10]

Compared with these approaches, YOLO based models perform detection in a single stage architecture, enabling much faster responding time. This makes YOLO particularly well suited for automated fish counting, where both speed and reliability are critical.

The F1 score combines precision and recall, balances between detecting fish and avoiding false alarms. The peak of the curve above (Figure 10) shows the optimal confidence threshold where precision and recall are balanced most effectively. This confidence threshold is implemented within the counting system code to determine whether the object should be counted as a fish.-During future field testing, additional training footage collected from the completed fish transfer system may be used to further refine the model. Although the numerical values of the evaluation curves may change after retraining with new data, the overall trend and optimization process are expected to remain consistent.

Transfer Unit Overview

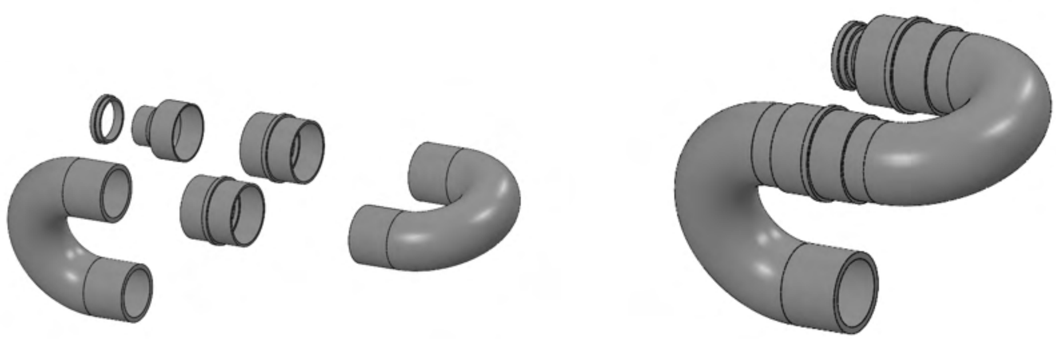


Figure 11: Exploded View of PVC Sub-Assembly

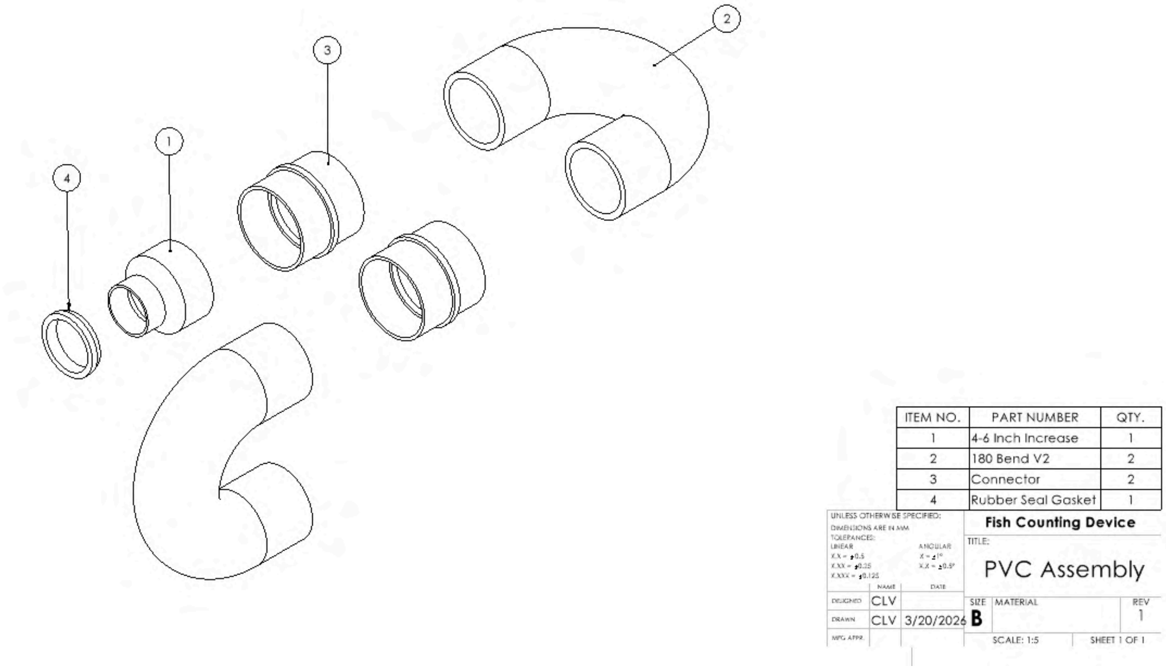


Figure 12: Drawing of PVC Sub-Assembly

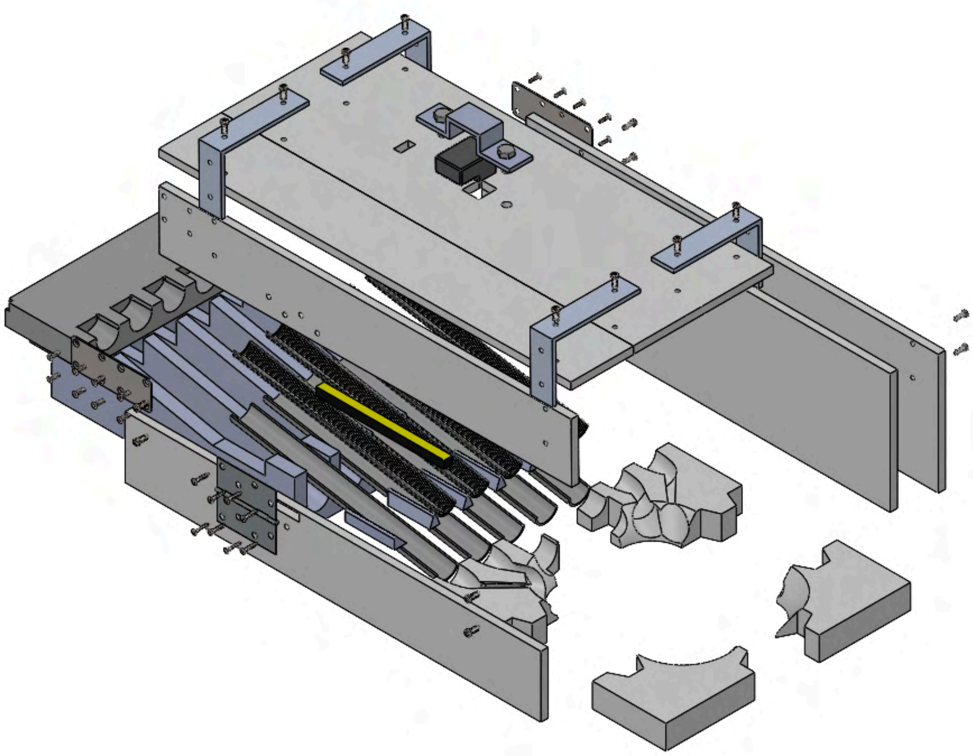


Figure 13: Exploded View of Grooves Unit Sub-Assembly

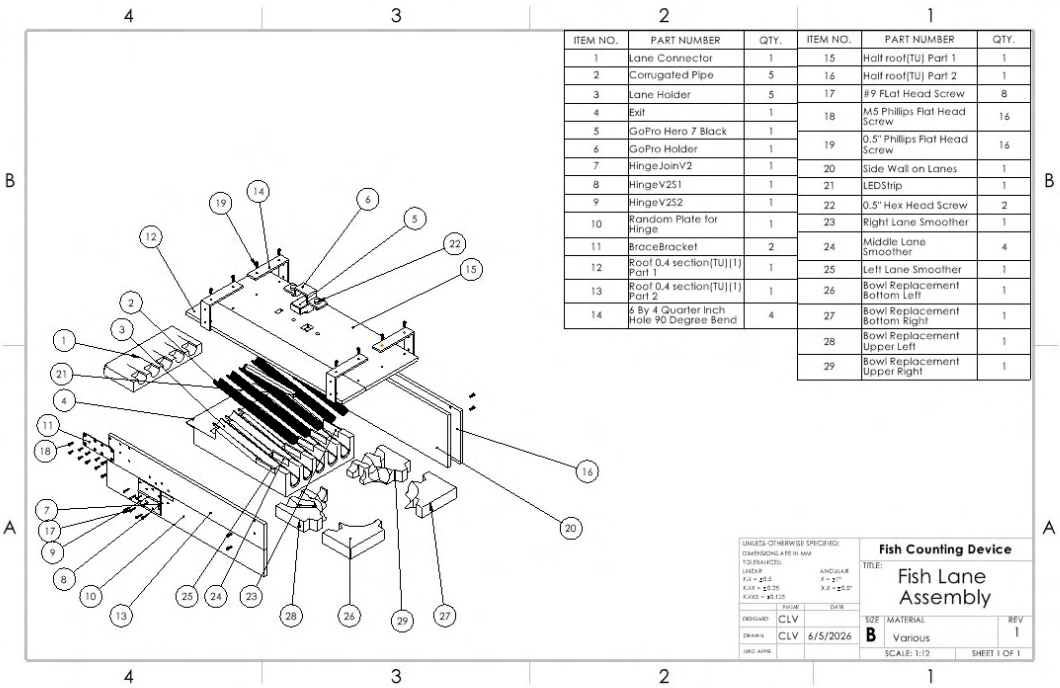


Figure 14: Drawing of Grooves Unit Sub-Assembly

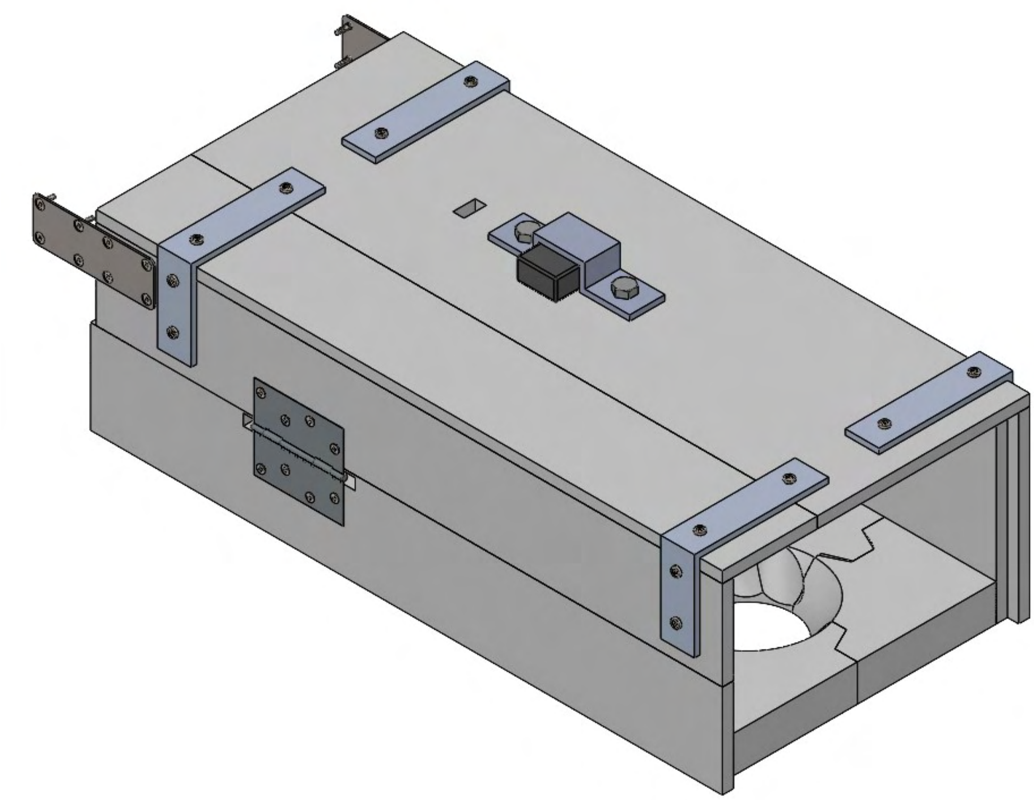


Figure 15: Isometric Closed View of Transfer Unit Sub-Assembly

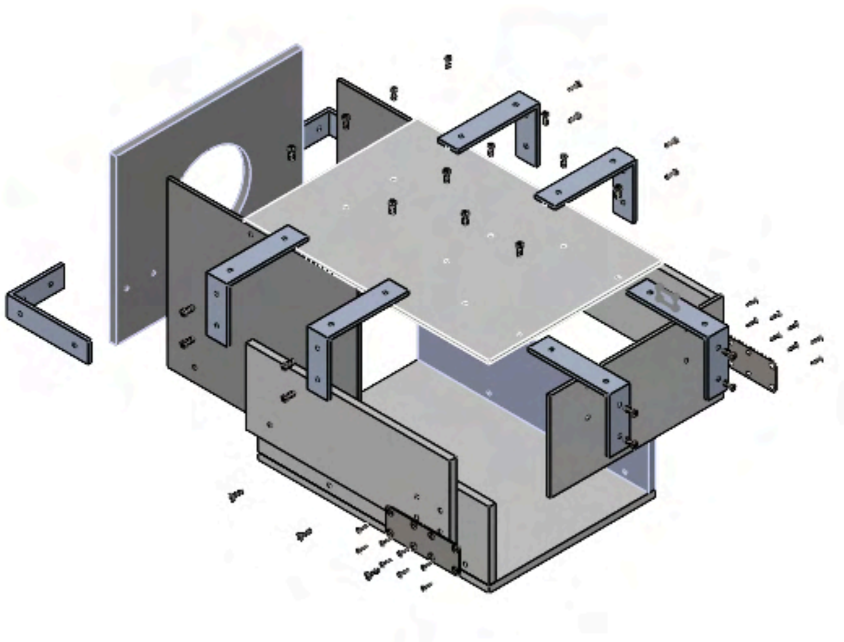


Figure 16: Exploded View of Transition Housing Sub-Assembly

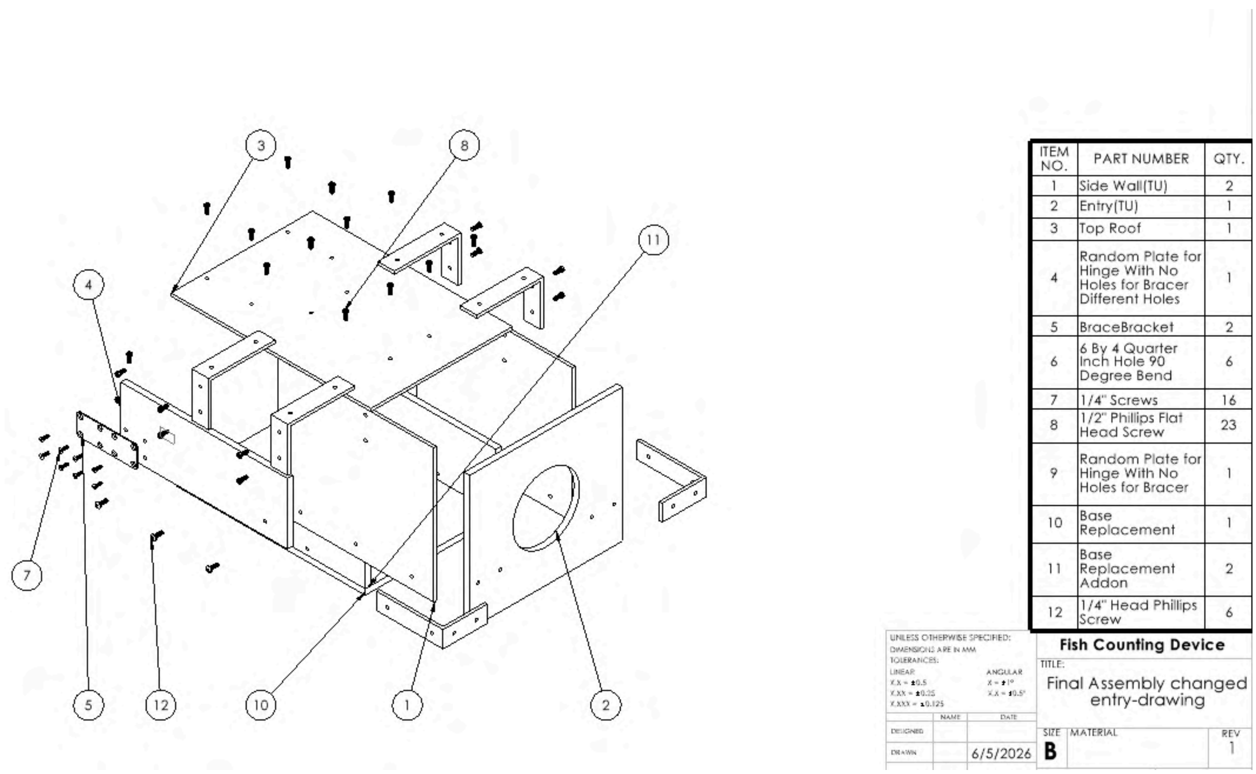


Figure 17: Drawing of Transition Housing Sub-Assembly

The transfer unit has been designed so that the 4" tube (connected to the pump) is connected to a 6" diameter PVC pipe with two 180° bends. The increase in diameter will slow the speed of the water down enough for the camera to capture the fish in video without blur (TUR. 1.2). The speed of water through the housing is less than 0.36 m/s (14 in/s). The PVC pipes are then connected to the housing device where the fish are separated into individual lanes to avoid overlap between multiple fish (TUR. 1.1). The transfer unit contains a cut where the GoPro can be placed onto from the top (TUR. 1.3). Due to the fixture that will be attached on top of the transfer unit, we made sure that the camera is securely placed, reducing the likelihood of it coming off. We assumed that the bracket and bolts are strong enough to withstand the vibrations of the pump but we are utilizing an interference fit to ensure that the camera lens does not shake.

The transfer unit subsystem is securely placed over the attachment unit. The wooden material provides facilitation for manufacturing and allows us to fit the four hooks onto the RV. The weight is dispersed onto two wooden hooks. We based the weight requirement based on an already existing fish counter that is able to track 5000 fish per minute while we are attempting 1000 per minute. The weight of their smallest model is about 130lbs so our goal is to be lighter than that [11]. The individual lanes that we are utilizing are called “grooves” (TUR 1.1). We are using them as a way to try to separate the fish to make it easier for the sensor subsystem to pick up how many fish there are. The grooves are made out of corrugated pipe. The corrugated pipe was mentioned by the research team as its safer for the fish and the shape of the pipes prevents the fish from hitting the edges when water is flowing through. The pipes are then cut in half and are placed as 5 different grooves. The size (radius) of the grooves was determined to be based on the average width of the delta smelt being half an inch [2]. The design idea revolved around the fish being unable to overlap with each other without moving onto another set of grooves (as we noticed from the testing footage that the research team sent that the fish often overlaps making it hard for them to count the fish).

One of our other requirements involved there being lighting for the sensor. We know from our sponsors that red light does not negatively impact the fish [2]. So we used two red-light strip lights that were simply connected to the computer through a USB. They are placed around the sensor to ensure enough visibility for the camera and computation (TUR. 1.5). Using another form of light might require an electrical outlet that would be slightly more challenging for the team to set up, granted they are on field and not in the laboratory. And using battery powered lights brings in the chance of it running out while the fish are being transported. The research team won’t have the opportunity or time to run the system again. Hence, why we used a USB connected light as it can already use the computer we are using and doesn't have the chance of turning off mid-transfer.

Attachment Overview

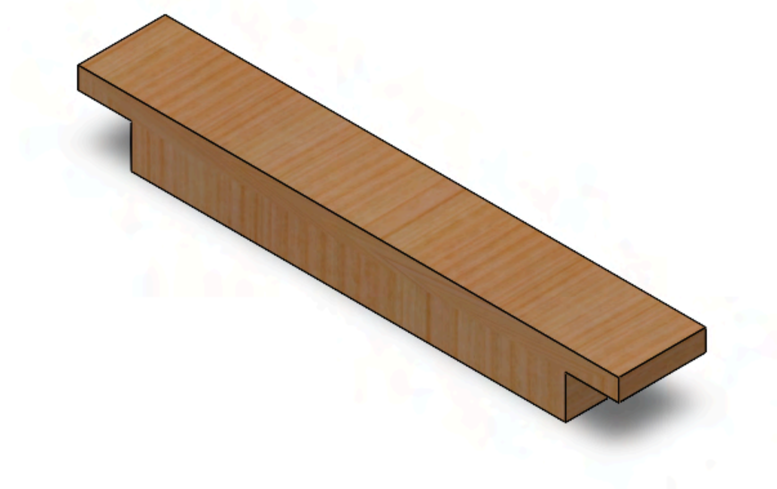


Figure 18: Wooden Beam Attachment Subsystem

The attachment unit is responsible for suspending the transfer unit on the RV. Two wooden beams have been cut to size so that they sit snug onto the RV, they are both about 50 inches in length. Both beams are placed so that the exit bowl of the transfer unit has enough space so that the fish may exit. These beams will disperse the force of the transfer unit (with water in it and with no water in it) to the 4 areas of contact at the end of the beams (AR. 1.0). The beams are light enough for one person to carry a beam at a time and since the beam is long length wise and short in height and width, this makes transporting them very simple since they can be placed on either side of the trailer (AR. 1.1). Each beam was proven to have a snug fit with the RV meaning that as the transfer unit is placed on top of the attachment subsystem, they will be able to withstand the water flow and will not be the cause of any shaking (AR. 1.2)(AR. 1.3).

Bill of Materials

Part Name	Link	Description	Fabrication	Material	Donated	Vendor	# of units used	Price per unit	#Packs to buy	Total Price	Total Cost Before Tax	Total Cost After Tax
PVC S&D Increaser/ Reducer Coupl	PVC Increaser	Coupling for 4-6 inch cori	FALSE	PVC	☐	Home Depot	1	11.74	1	$11.74		
PVC DWV Return Bend (6 inch)	Pumbing Sell Return Bend	180 Degree Return Bend	ordered	PVC	☐	Plumbing Sell	2	68.56	2	$137.12		
PVC 6 Inch Tube Coupling	6 in Tube Coupling	Coupling for the 6 inch tu	FALSE	PVC	☐	Lowe's	1	8.98	1	$8.98		
GoPro Hero 7 Black Camera	GoPro Hero 7	Camera that is used for s	-	Multiple	☑	Amazon	1	359.99	1	$359.99		
LED lights	LED lights	Red lights	ordered	-	☐	Amazon	1	5.09	1	5.09		
PVC Primer and Cement	PVC Primer and Cement	Gluing PVC parts togethe	FALSE	-	☐	Home Depot	1	11.98	1	11.98		
1/4" PVC Sheet	1/4" PVC	24" x 48" PVC sheet, Use	ordered	PVC	☐	BuyPlastic.com	1	53.58	1	53.58		
1/2" PVC Sheet	1/2" PVC	24" x 48" PVC sheet, Use	ordered	PVC	☐	BuyPlastic.com	2	104.78	2	209.56		
1/2" PVC Type 1 Plastic Sheet	PVC Chemical Resistant Plastic Sheet	24" x 48" PVC sheet, Use	ordered	PVC	☐	BuyPlastic.com	1	130.32	1	130.32		
Door Hinge	Door Hinge	4 in. Square Radius Chro	FALSE	Steel	☐	Home Depot	1	4.93	2	9.86		
Metal Bracket	Metal Bracket	5.4" × 1.9" × 0.08" , Com	ordered	Steel	☐	Amazon	2	12	1	12		
Corrguated pipe	Corrugated Pipe	1 in diameter x 20 feet lo	ordered	PVC	☐	Amazon	1	27.56	1	27.56		
6x4" L shape bracket	L Bracket	6x4 Inch Stainless Steel	ordered	Metal alloy	☐	Amazon	10	19	3	57		
Capture Card	Capture Card	Display camera video ont	ordered	-	☐	Amazon	1	27.99	1	27.99		
4' Rubber Gasket	4" Camlock Gaskets	Seal for water	ordered	Rubber	☐	Amazon	2	7	1	7		
Mini HDMI to HDMI	Mini HDMI to HDMI Cable	Used to plug to capture c	ordered	-	☐	Amazon	1	9	1	9		
Micro HDMI to HDMI	Micro HDMI	connect camera video to	ordered	-	☐	Amazon	1	6.18	1	6.18		
Aluminium Rectangle Bars	aluminum-rectangle-bar	Aluminium at the end of t	ordered	Aluminium 6061	☐	OnlineMetals	1	140.44	1	140.44		
PLA Filament	PLA Filament	PLA Filament for grooves	ordered	PLA	☐	MatterHackers	6	19.86	6	119.16		
Wood	Wood	Wood for attachments	TRUE	Douglas Fir Woo	☐	Knudson Lumbe	2	64.66	2	129.32		
					☐						**Before Tax:**	**After Tax(9.25%)**
					☐						Total Without Donated Parts	
					☐						1473.87	1610.20
					☐						Total With Donation	
					☐						1113.88	1216.91

SECTION 4: RISK ASSESSMENT AND ANALYSES

The team conducted a Design Failure Mode and Effects Analysis (DFMEA) in order to determine what would be the most likely points of failure and how severe would they be towards the completion of our project. The overall DFMEA can be found in the appendix (). Once the probability of failure, severity, and detection were tallied, we decided that anything above a 60 would be considered high risk and would need to be further evaluated. Most of the high risk areas were dependent on if the fish would die, potentially cause damage to the overall system, or create some inaccuracy in our final count.

Prioritized List of Risks

Part at Risk	Potential Effect of Failure	Risk Priority Number
Entry Piping (Includes 180 Bends and any Connectors)*	Might not be fastened in enough and could come loose with the PVC cement we are using.	112
Entrance of Transfer Unit	Might not be fastened in enough and could come loose	64
Hinge	Might break under weight of the Roof 0.4 Section Assembly	90

Table 2: Simplified DFMEA - Transfer Unit

Part at Risk	Potential Effect of Failure	Risk Priority Number
L Brackets and Brace Brackets*	Transition between subsystems may not be smooth enough or can cause damage to the final system.	280
Go Pro*	The electricity in an exposed wire touches water	160

Table 3: Simplified DFMEA - Sensor

Part at Risk	Potential Effect of Failure	Risk Priority Number
Stabilizing Block	Slips or can not handle the weight of the final product and results in the whole system collapsing into the final tank.	150
Stretcher*	Can not hold the weight of the transfer unit and collapses into the tank.	200

Table 4: Simplified DFMEA - Attachment

**high priority risk*

High Priority Risk Management Strategy

Entry Piping

Problem Statement

The entry piping plays a major role in reducing the speed at which the delta smelt enters the transfer unit. It is essentially designed to reduce the speed of the fish to prevent any sort of harm. Any leakage, turbulence or sudden change in the pump speed can affect the fish's safety. To ensure stability, it is important that there are no leaks and that there is no slipping or lack of friction between the connections.

Assumptions, Principles, Methods

1. The piping will made out of PVC
2. There will be multiple pipes used in combination
3. The PVC cement and primer will be used as a sealant

Observed Failure Mode

There were gaps in the entry piping causing leakage of water, which would disrupt the water flow for the minnows. Moreover, the pipes would bend downwards due to the flow of the water and its weight causing the water to spill back and accompany the minnows' resistance to the flow. This resulted in some minnows falling out of the piping system.

Design Modification

We added wooden supports that attach the bottom of the entrance piping to the top of the entrance to the cuboidal unit. We did this by utilizing Titebond wood glue. This allowed for more support to the pipes as the force of the water would bend it and the weight of the pipes would naturally bend it irregularly. We attempted to fix the leakage from the piping by adding PVC cement and primer between the different pipes.

Conclusion

It provided more reaction support to the pipes and prevented it from bending. However, there was a small tilt on the pipes that would spill water back to the user. This also pushed the minnows back, although the fish falling back onto the user is partially due to minnows resisting the flow and swimming against it. The sealant that we used to connect the piping did reduce the leakage from the entry piping but not entirely. But it was minimal and would not pose an issue to the hydraulics lab during testing.

L Brackets and Brace Brackets

Problem Statement

The L Brackets and Brace Brackets are an important component of the entire assembly as they connect different parts together. It is the main holding component of the product. It is important that there is not immense force on the brackets that it may break causing damage to the connecting parts and failure of the assembly. To prevent failure, the brackets should experience adequate force without causing deformation to the material itself.

Assumptions, Principles, Methods

1. The brackets should not bow down or snap when applied onto the transfer unit
2. The movement of the brackets should not damage the

Observed Failure Mode

The two brackets did not face any design failure apart from the distance of our drilled holes.

Design Modification

We drilled new holes so that it matched the amount of screws we believed were required to maintain the stability of our project.

Conclusion

The system is now more stable with no risk of it snapping under its own weight.

GoPro

Problem Statement

The GoPro holds great importance in calculating the number of fish passing through the system. It is possible, with a water environment, that some exposed wires may come in contact with the water causing a short circuit and potentially damaging the camera. To prevent this, the connecting cables should be insulated properly and for further assurance be assigned to a designated path away from water flow and splashes. Moreover, we need to ensure that the camera is able to view the fish passing through well enough for the program to recognize and track the delta smelt.

Assumptions, Principles, Methods

1. When the program starts, the GoPro will be on already.
2. When the program starts, the GoPro lens will be facing the fish
3. It should be able to run at the FPS required

Observed Failure Mode

Some water did accumulate on the lens of the camera when the system was undergoing a lot of water.

Design Modification

We decided to reduce how far the camera was protruding to see into the lanes by adding more material on the flat side of the camera which pushed it up.

Conclusion

The added material pushed the camera up far enough to reduce the amount of water on the lens, but still provides a clear view of the lanes. This change will reduce the likelihood of something going wrong with the GoPro feed.

Stretcher

Problem Statement

The stretcher is the part that holds the entire system above the tank as the fish pass through it. Potential risk of failure involves deformation of the part and causing the system to fall into the tank. To prevent that, it is important to ensure the piece locks properly and does not slip as water flows through the transfer unit. The piece must latch onto the RV tank well enough to support the weight of the transfer unit. Slipping is the most likely form of failure.

Assumptions, Principles, Methods

1. The transfer unit will be placed over the tank provided by the hydraulics lab.
2. The wooden planks will be held onto the tank over the length of the tank and transfer unit.
3. The wood utilized with be Douglas Fir Wood

Observed Failure Mode

Our observed failure mode was the inability to properly attach to the fish tank hence increasing the likelihood of slipping. The hooks of the wooden plank couldn't fit on both ends, which causes the transfer unit to be titled forward and can cause issues for testing with waterflow coming out.

Design Modification

Our design modification was to return to the manufacturing stage and cut off the side of the fitting edge that attaches onto the tank so that it would properly fit.

Conclusion

The wooden planks were cut down and successfully attached onto the tanks. We also know that the weight of the transfer unit can be placed onto the woods without it breaking as previously calculated with the safety factor of the wood.

SECTION 5: PROTOTYPE TESTING PLAN

Test 1: Video Camera Testing

Objective:

We will be seeing what is visible and detectable in the current set up of the camera and lights when the code is running.

Test Design:

The entire prototype will be tested at the same time as the program must run under the same conditions it would in the nontesting environment. We will need an external computer to run the code and the pump in order to cycle water through the system. We will use minnows in order to mimic the fish swimming through the system as the sponsors have stated that the minnows are similar in size to delta smelt and are considered "feeder fish", and are thus expendable. The minnows share a very similar appearance to the delta smelt which helps in fine-tuning the model based on the actual fish. Importantly, the fish also have slight reflective properties like that of the delta smelt. We will collect the video of the water and fish passing through the lanes. After recording the video, we will run the video in the program to see the count of the fish and compare the number of fish passed through the system and compare the accuracy of the model.

Data Collection:

Test Case	Fish Transferred	Fish Counted By AI	Accuracy (%)
Test 1	79	51	64.6%
Test 2	48	81	31.3%
Test 3	30	25	83.3%

Table 5: Accuracy of Fish Counted

Data Analysis:
From the 3 tests we did, we had the highest accuracy for the third test, which had the lowest number of fish pass through. For test 2, we noticed a few fish got stuck between the lanes and due to the fish flopping around, the AI counted the fish multiple times counting each fish more than once and exaggerating the final count by a factor greater than 1. To overcome this, the model needs to be trained more so that it recognizes that it is still the same fish even when it is flopping around. During test 1, it was observed that the fish were moving at high speed which the model could not identify, even at a higher frame rate resulting in the recorded accuracy. On average, we have an accuracy of 60%. This accuracy is too low, but should only increase as the model is trained more. Unlike test 2 and 3, test 1 had close-to-average accuracy.

Safety Considerations:
A safety consideration is slipping as there will be water in the environment. To prevent this, the test will be conducted outside and people will stand at both ends of the housing, minimizing risk of the water coming into contact with people.

Limitation:
We are assuming that the minnows will flow properly and that they will go through the lanes appropriately. For the camera, we also will need to use an external method to capture the video on the computer as the code itself does not have a method to record video. In this the quality may suffer and muddy results.

Test 2: Leak Test

*Objective***:**
We want to see if the container is water tight and analyze how much water releases from the container. We want this as water leakage will affect the flow of the water and change how fish flow, which would affect our results.

Test Design:

We will suspend the transfer unit in the air, placing sawhorses on both sides to hold it up. We will then run the pump for 5 minutes once it has achieved steady flow with water being collected in the RV tank. After the 5 minutes are up, we turn off the pump, letting any water naturally flow out of the system into the tank.

Data Collection:
Running the test twice for a total of 10 minutes allowed us to really see the extent to which the water was flowing out of the system. We could see two full cycles of the water surging and waning. From this we could see obvious little leakage that resulted in constant small streams stemming mostly from the lanes section of the system. In the piping section, we could see drops leak from the belly of the connection between the two 180° bends, as can be seen in Figure 19 below. Additionally, there was a pair of finger sized holes in the conversion from the flat area to the lanes.

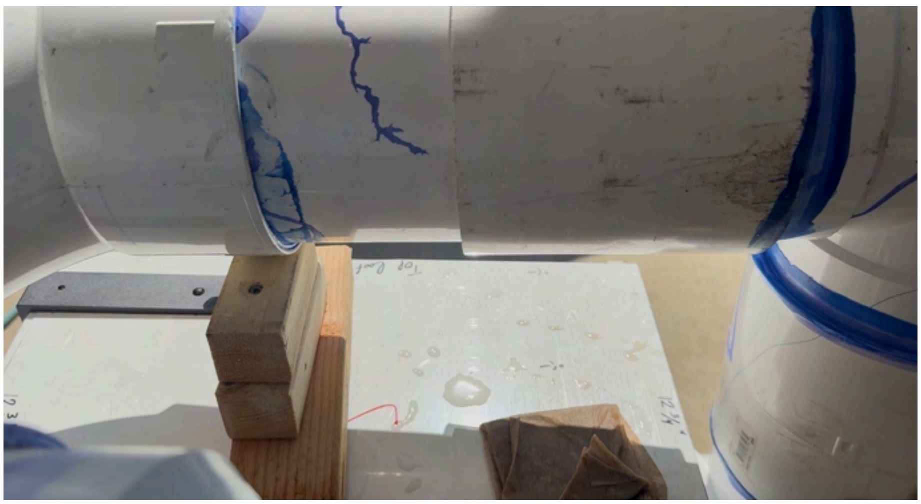


Figure 19: Leak Test on Piping Section

Data Analysis:
From the first test, there was heavy leakage from different parts of the transfer unit resulting in the system breaking down from underneath. There was heavy leakage in the first run due to this limitation. Moreover, there were many unsealed spots through which water would escape. To patch the holes coming in the lanes section, we applied silicone all along the edges of the walls and in the lanes. For the belly of the piping connections, we applied more PVC primer and cement onto it as the silicone was too big to enter the small problem area. To patch the two big holes, we put a small, thin rod of leftover PVC sheet and placed it covering the problem areas. Then we applied silicone on the inside of the system, as well as on the outside in order to hold it in place and to cover the hole up on both sides. Our ideal goal was to have no leakage, however with our design it is more practical for manufacturing that the gaps are not big enough for any of the fish to escape through.

Safety Considerations:
A safety consideration is slipping as there will be water in the environment. To prevent this, the test will be conducted outside and in an area where there is adequate drainage.

Limitation:
A limitation could be that we don't test the higher areas of the assembly so we don’t check if the whole thing is waterproof. Thus if the lpm of the pump was ever increased, then some more parts could be revealed as non waterproof.

Test 3: Stability Test

Objective:
We will see what parts come loose or are displaced greatly when the system is shaken under constant cyclic motion.

Test Design:
The entire assembly will be on two sawhorses where one is placed in front and the other in the back. We will position one member in front of the assembly and another in the back. The team members will take turns pushing the assembly forward by simply extending their arms forward, while gripping the system, in increasingly quicker time increments starting from every 5 seconds, then decreasing 1 second until every second after each member has individually pushed it 5 times. This will be repeated horizontally as well.

Data Collection:

Trial	Any visible detachments?(Y/N)	What happened
1	Y	The bottom of the transfer unit fell off (grooved lanes)
2	Y	Large bend in entry piping due to its own weight
3	N	The unit held together (slight bend in piping)
4	N	The unit held together (slight bend in piping)
5	N	The unit held together (slight bend in piping)

Table 6: Stability Test Results

Figure 20: Entry Piping Tilt

Data Analysis:
From the first two trials, there were a few design changes that were made to improve the stability and binding. After the first run, the floor of the transfer unit was not adhered tightly and the gaps resulted in the system floor coming off. To make up for it, a PVC sheet from the stock was used to accommodate the shortcoming, and provide support for all the parts inside. The additional layer of PVC on the bottom also allowed us to meet one of our original goals of making sure that it is easily manageable for the hydraulics lab to move and implement on field. It became easier as two people can hold onto a single sheet of PVC rather than the intricate corners of our transfer unit. From the second run, the entry pipe was observed to be bending which increased the stress around the curves. To solve this issue, a stack of wood was glued on to align the pipes horizontal and parallel to the transfer unit. This fix would also prevent water from splashing back to the user.

Safety Considerations:
It is possible that an object could fly off or into a member. For that reason everyone will wear standard PPE including safety glasses, closed toe shoes, and long pants. Moreover the 3D printed parts are relatively sharp and should be handled carefully.

Limitation:
We didn't have the ability to use the actual pump, so we are unsure on whether the strength of the vibrations from the pump will worsen the footage or cause the piping to lower more once again. The piping still requires slightly more stability as we noticed from our tests that it can leak water from the outside when we don't utilize a pump.

Test 4: Fish Mortality Rate Test

Objective:
The objective of this test is to understand the overall safety of the fish and the total mortality rate when the fish travel through our system.

Test Design:

A sample size of 100 minnows will be used in this experiment, and they will be passed through the system. Each time the fish are run through, the number of fish severely exhausted or nonresponsive to stimuli will be counted and transferred to a separate container for their safety. The fish that appear to be healthy will be passed through to the system after an interval of 15 mins in order to provide rest and reduce stress on them. This test will be repeated 3 times in order to understand the safety of the fish without exposing them excessively. The fish who seem exhausted will be considered in the mortality rate as the minnows are stronger than the delta smelt and have more endurance comparatively.

Data Collection:

Test No.	Input fish (x_1)	Healthy fish from the exit (x_2)	Mortality Rate($\frac{x_2 - x_1}{x_1}$)
1.	98	98	0%
2.	98	95	3.06%
3.	95	91	4.21%

Table 7: Mortality Rate from System Testing

Data Analysis:
From our tests, we have an average mortality rate of 2.43% which is less than the expected mortality rate from our sponsors. The fish are put through high stress from each run. Consequently their health is affected after each test resulting in higher mortality rate progressively. Ideally, each test should be done with a different batch, but to maximize efficiency with the time constraint, such design was ideal. Moreover, the minnows tried to resist the flow and tried swimming against it which the delta smelt do not follow, therefore the fish may be safer comparatively based on this assumption.

Limitation:
The minnows are much stronger and can endure more than the delta smelt, therefore the result may not give a completely accurate understanding of the fish safety when the fish pass through the system. And the minnows are darker than the delta smelt, and hence would be easier to see on the GoPro footage.

SECTION 6: CONCLUSIONS, LIMITATIONS, AND FUTURE WORK

Strengths

The strengths of this system is that it is rigid and does not break under its own weight. It can be handled fairly roughly without it breaking. The waterflow, despite us not being able to use the pump itself, functioned as we expected. The water landed in the area we designed for the fish to arrive in and the water was able to push them through the grooves (TUR 1.1). Moreover, after multiple tests, we can conclude that the grooves do help split up the fish well enough to make machine learning analysis easier. The transfer unit also is compatible with the system that the Hydraulics lab already utilizes. The lighting was enough for the camera to see the fish, and with the red lights that we used that are also easily connectable to the computer, it provides more lighting for the unit in case the lab conducts their research on days that are cloudier and have less natural lighting. It is steadily powered (SR.1.0). The GoPro was extremely stable and firmly in place throughout the experiment. We tested removing the bolts that firmly secured the GoPro and noticed how it's still able to record the fish well enough and without noticeable shaking.

Weaknesses

The design that ended up being the final product is too large and heavy to be quickly deployed and needs two people in order to properly lift it. Additionally, the first part of the system with piping is too heavy and required emergency support in order to hold itself up without putting strain on the entry plate. We found that there is still leakage that occurs throughout the transfer unit. The leakage was not enough to allow the fish to escape the system but, based on our original goals, we would need to add additional sealants or manufacture more precise pieces to fill in the gaps. Moreover, we noticed how there was a small enough gap between the lanes and the groove PLA structure that certain fish could get stuck in. They were stuck in a way in which the water flow couldn't push them out and required someone to fish them out into the exit hole. Although we were able to capture the 5 lanes to use for the machine learning, having the camera more towards the center would have allowed for a more consistent viewing of all the lanes. Since the piping wasn't completely smooth, some of the fish may have been forced to deal with additional stress. Most importantly, since we couldn't use the pump that the facility actually uses for the delta smelt, we can't confirm definitively that the water flow and speed is adjusted correctly for the camera to see the fish. The machine learning program was also able to produce a counting error of 82.467% with more than 5000 manually annotated and over 10,000 online data training. As the users keep collecting real data samples, they can train the model to be more accurate for their real operation environment.

Future Work

Future work should prioritize reducing the weight of the unit in order to allow it to be potentially carried by one person, or at least make the assembly easier to disassemble in order to allow it to be stored easily. This could include choosing a lighter, yet still fish safe, material that would naturally reduce the weight even if nothing else is changed.

Finding a quicker way to reduce the fish speed would also allow for the weight to be reduced later on by potentially removing the middle portion of the assembly, leaving the lanes and computer vision to operate as is.

More importantly, tightly sealing the gaps between the edges of the lanes to avoid fish getting trapped in the small gaps. Although something that should absolutely be done is to continue training the algorithm. Training the model furthermore based on real footage will fine-tune the algorithm to count the fish correctly.

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APPENDIX

Appendix I: Manufacturing/Production Information

Fabrication Process

PVC Sheet Parts

We used three ½" thick and one ¼" thick 24" x 48" PVC Sheets to design the parts from PVC. We outlined an optimized outline for all the walls and roof based on the drawing package, and cut them using a handheld bandsaw by clamping part of the sheet to a table for stability. After cutting each part, the edges were sanded and smoothened to remove irregularities while connecting the parts together.

3D Printed Parts

The lanes, lane connector, GoPro holder, and the entry to the transfer unit, were all 3D printed using a PLA Filament on the Bambu P1S 3D Printers, using 3 wall loops and 25% infill for low density but high strength, and at a nozzle temperature of 220 ℃. The lanes were too big for the printer and required to be broken down into 4 quadrants and were then assembled using an adhesive. They were split in SOLIDWORKS so that each piece could be printed within a build space of 256 x 256 x 250 mm^3.

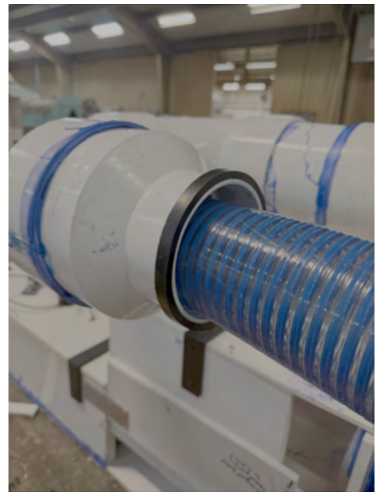
Corrugated Pipes and PVC Pipe Lanes

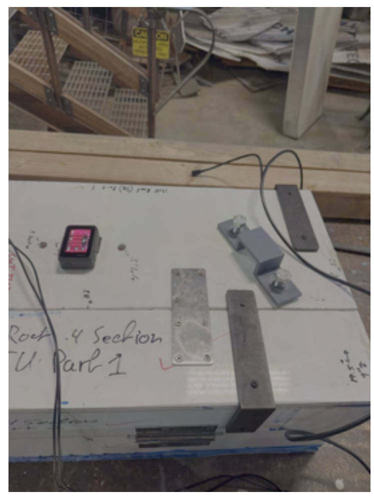
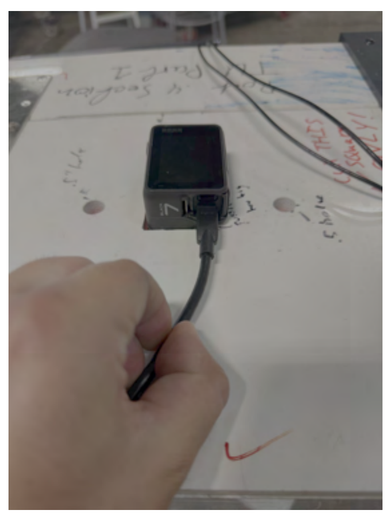
The corrugated pipes were bought and modified by cutting 5 pieces of 14" length. Then each of the pipes were cut along the length from any one point on the circumference, trying to be as straight as possible to avoid irregularities. These corrugated pipes were glued to the lanes that were 3D Printed to provide a surface for the fish through according to their comfort. It is necessary to wear a PPE kit with a mask covering the mouth and nose when cutting the PVC pipes and sheets to prevent breathing the dust.

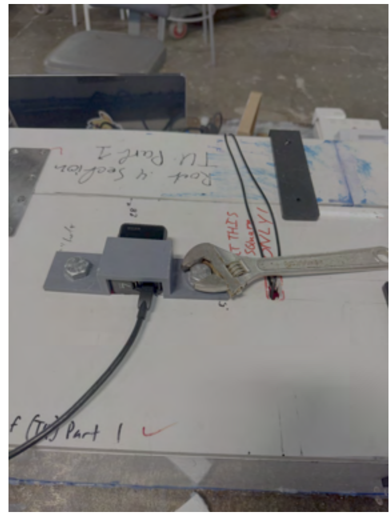
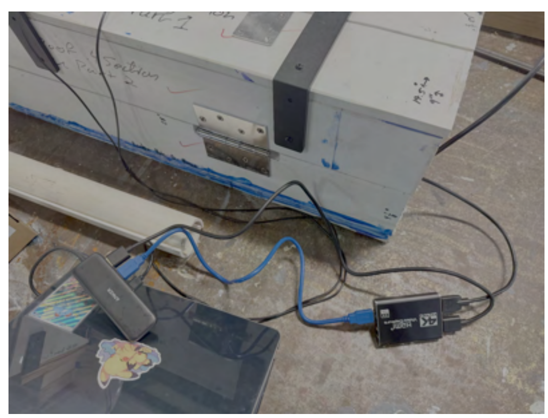
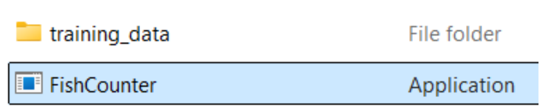
Quick User Guide

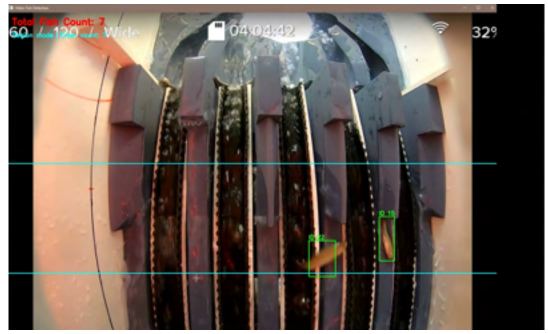
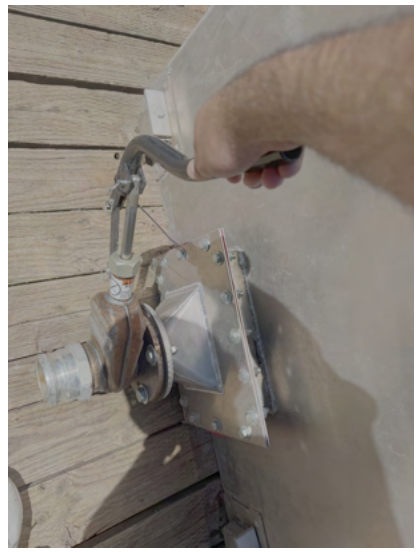
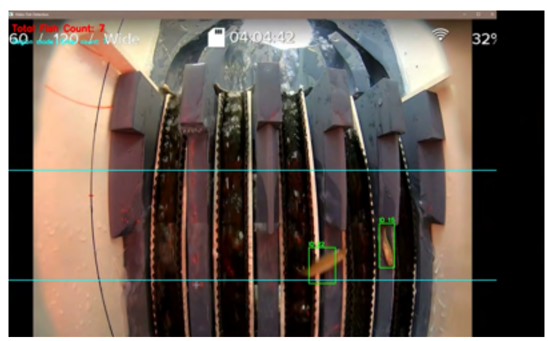
This solution was initially designed with the idea that it would go on top of the tank, depositing fish into it. During testing, it was discovered that it could be operated after the tank, where the only change would be that the outlet of the hose needs to be held at the inlet of the system manually. As such, the mounting of the device is not listed here as it may not be necessary depending on the current situation and need they find themselves trying to satisfy. Additionally, the code to run the fish counter was exported as an exe file, where the instructions on how to download and operate the programs can be found below the quick user guide.

Step	Instructions	Photo

1:	Place the hose into the inlet of the system	
2:	Turn on GoPro	
3:	Set GoPro to video mode	

4:	Make sure GoPro is in 960|240 resolution|frames per second mode and wide field of view	
5:	Place GoPro with top of GoPro to the outlet	
6:	Connect micro HDMI to HDMI cable to left side of GoPro	

7:	Fasten down GoPro with GoPro holder	
8:	Connect HDMI cable to input of capture card	
9:	Connect blue power cable and output HDMI cable to computer along with LED power cable	
10:	Turn on code	

11:	Press S to begin the code counting	
12:	Turn on water	
13:	Count fish	

14:	Turn off water	
15:	Press Q to stop code	

File Download Instruction

- Download the FishCounter exe file into a folder
- Create a Training Data folder at the same level as the exe file
- Download the best.pt file into the Training Data folder
- Open the exe file by double clicking it
- Press S to start counting
- Press P to stop counting
- Press R to reset counting
- Press Q to exit the program

Appendix II: Project Management Information

Item #	Subsystem / Component Description	Component Function	Potential Failure Mode	Potential Effect of Failure	Severity of Effect (1-10)	Potential Cause of Failure / Failure mechanism	Occurrence: Probability of Failure Cause (1-10)	Control Method (Ability to detect & mitigate, or limit exposure)	Detection	Risk assesment = RPN = Sev x Occur x Detect	Recommended Action	Action Results (New RPN)
				1 = annoyance, have to fix afterward 3 = uncomfortable, shut it down, loss of time, loss of small $ 6 = loss of significant time, significant damage to components 8 = huge loss of time, $, equipment 10 = Any injury		1 = cannot think of any way 2 = designed to prevent with all known uses / customers 4 = Unlikely 6 = Rare, but have seen it happen 10 = Very likely to happen in a single test		1 = Well designed control will work every time 3 = Likely detection (slow failure + good signals) 5 = Good detection, (moderate fail rate + ok signals) 8 = Some chance to detect (fast fail + human reactions) 10 = No chance to detect until too late				
1A	Entry Piping (Includes 180 Bends and any Connectors)	Increaser fluid diameter from 4 inch pump to 6 inch hose and to increase time in the system to allow for fish speed to reduce and facilitate camera visuals.	Might not be fastened in enough and could come loose with the PVC cement we are using.	Disrupts flow, fish would be out of water	8	Slipping/ lack of friction	7	Easy to detect and during assembly we will see if it is loose	2	112	Return to manufacturing process or use better adhesion to combine the pieces	
1B	Entrance of Transfer Unit	Includes the hole in the wall for the flow thats connects to the rest of the system as well the walls of the transfer unit entrance	Might not be fastened in enough and could come loose	Disrupts flow, fish would be out of water	8	Slipping/ lack of friction	2	Easy to detect and will be able to check during manufacturing is something is wrong	4	64	Return to manufacturing process or use better adhesion to combine the pieces	
1F	Rubber Seal Gasket	Creates seal from pump to increaser	Does not correctly or adequately seal	Water or fish could leak	3	Incorrect application	6	Easy to detect and will be able to check during manufacturing is something is wrong	3	54	Order correct seal size or find strong adhesion	
2A	PVC Sheets used for walls and roof of transfer unit	Acts as a roof and walls to block the sun	Not flush with walls	Sun comes through	9	Slipping/Lack of friction	1	Easy to detect and will be able to check during manufacturing is something is wrong	1	9	Return to manufacturing process or use better adhesion to combine the pieces	
2B	PVC Base Plates	Wall that also includes the floor that connets to the next part, fish lose friction here	Could not be flush with the wall or floor and then create a leak	Fish or water could leak through	8	Slipping/lack of friction	1	Easy to detect and will be able to check during manufacturing is something is wrong	3	24	Return to manufacturing process or use better adhesion to combine the pieces	
2C	Random Plate for Hinge with No Holes for Bracer	Lines up walls so that they are connected with bracer	Not screwed in properly and they fall off	Transition between subsystems not smooth	1	Manufacturing failure	5	Easy to detect and will be able to check during manufacturing is something is wrong	1	5	Return to manufacturing process	
3A	Half Roof Transfer Unit	Is a roof with holes in it to allow the go pro access	Sides are not flush with other roof and walls	Light gets through	1	Material selection error and slipping	2	Easy to detect and will be able to check during manufacturing if something is wrong	3	6	Return to manufacturing process or use better adhesion to combine the pieces	
3B	Roof Transfer Unit	A roof that has a hinge attached to view inside	Sides are not flush with other roof and walls	Light gets through	1	Slipping/lack of friction	2	Easy to detect and will be able to check during manufacturing if something is wrong	3	6	Return to manufacturing process or use better adhesion to combine the pieces	
3C	Hinge	A hinge to allow for the closing and opening of the viewing area	Might break under weight of the Roof 0.4 Section Assembly	Roof 0.4 Section breaks off and allows for light to pass through	6	Miscalculation of force applied	5	Would be audible noise and visible signs that the hinge is succumbing to noise	3	90	Select better material or redesign hinge	
4A	Go Pro	To see the fish	The electricity in an exposed wire touches water	Go pro fails	8	Incorrect handling	4	Easy to detect and will be able to check during manufacturing if something is wrong	5	160	Reconsider camera selection and water proof safety	
4B	Go Pro Holder	To hold the Go Pro in place	Not flush with the roof or go pro	Go Pro is shaky	3	Manufacturing failure	2	Easy to detect and will be able to check during manufacturing if something is wrong	1	6	Return to manufacturing process or use better adhesion to combine the pieces	
5A	Lane Connector	Creates Lanes for pipes from the previous housing segment	Lanes are manufactured to wrong size	Corrugated pipes are too squished or the spacing is not enough for the intended 5 lanes.	8	Manufacturing failure	2	Easy to detect and will be able to check during manufacturing if something is wrong	3	48	Return to manufacturing process or use better adhesion to combine the pieces	
5B	Corrugated Pipe	The fish go through here creating a flow similar to the pump	Might not be deep enough	Fish and water might pour over	6	Manufacturing failure	2	Easy to detect and will be able to check during manufacturing if something is wrong	2	24	Return to manufacturing process or different piping product	
5C	PVC and Corrugated Pipe Connection	The fish go through here creating a flow similar to the pump	Corrugated Pipe does not stick to the pvc pipe beneath it well because the flow of water is too strong.	Flow is disrupted, camera vision is off, potentiaL harm to fish	7	Manufacturing failure	4	Hard to detect as we will not have the water on until we start testing	8	224	Apply some sort of adhesion to the top of the pvc tube and bottom of the pipe in order to hold them together better	24
5D	Exit	Sturdy structure to connect all the laned together			1	Manufacturing failure	2	Easy to detect and will be able to check during manufacturing if something is wrong	3	6	Return to manufacturing process in order to ensure likelihood of the exit capturing all the fish	
5E	Bowl Replacement	Pump pipe connects here and is where the fish end up	Edges are not aligned to walls	Water and fish could leak through	1	Manufacturing failure	2	Easy to detect and will be able to check during manufacturing if something is wrong	3	6		
6A	Stretcher	Grips the rv tank and holds up the counter assembly	Piece does not lock properly or slips	Counter slips	8	Material and slipping/lack of friction failure	5	Would be easy to see the the counter assembly start to move and we could stop it before something happens	5	200	Put some sort of adhesion on the connecting surfaces to lock in into place but still allows for removal	80
7A	Stabilizing Block	Touches the bottom of base replacement	Slips	Counter slips	6	Manufacturing failure	5	Would be easy to see the the counter assembly start to move and we could stop it before something happens	5	150	Put some sort of adhesion on the connecting surfaces to lock in into place but still allows for removal	80
8A	Lighting	Lighting to allow for camera to see all the fish coming through	Comes off transfer unit roof or isn't strong enough to see fish	Camera can't pick up on fish coming through	3	Product selection failure	5	Would be easy to see once we have the box fully set up. However, we would need to use a fish of similar colour, size and transparency to confirm the light being strong enough	5	75	Purchase new lighting if not strong enough and use a different adhesion technique if the connection isn't strong enough.	
9A	Brace Bracket	Lines up holes for screws	Too much force applied that they break	Transition between subsystems may not be smooth enough or can cause damage to the final system.	8	Miscalculation	7	This would be harder to detect as it could be a sudden quick failure one water is added to the system during testing	5	280	Add another bracer to split the force	120

Changes in Project Timeline

Our team originally planned for the arrival of our materials shortly after Spring Break ended. From here, we planned to begin construction of the Transfer Unit and Attachment subsystems, test the Sensor subsystem, and test the Computation subsystem. All three tasks would each be done at the same time over the course of weeks with construction being completed first, the sensor testing next, and computation testing last. From the tests we would be able to analyze the system overall and the amount of fish that have died going through the system. We would continue testing the system but work to finish the solution prototype.

Although, when we requested to purchase our materials through our sponsor, they were unable to get back to us until Spring Break finished. This resulted in the material purchases one week late and we received the materials we needed on week 3 of the spring quarter. The computation subsystem was being trained over this time and we worked to make it better and ready for system testing. However, when manufacturing the transfer unit, material orders got canceled or did not fit how they were supposed to, resulting in further delays to the completed transfer unit subsystem. What resulted was a completed system on week 7 of the spring quarter and two test sessions. The system was able to transport and count fish as expected.

Task List

#	**Task Description**
	Winter Quarter
	Understanding the Problem:
1	Prepare Questions for Sponsor Meeting
2	Memo Report 1
3	1st Sponsor Meeting
4	Sensor Research
5	Brainstorm Solutions
6	Find Solutions already being used
7	1st Group Meeting
8	Memo Report 2
	Subsystem Identification:
9	Sensors on the Market
10	Systems Currently in Use

11	Fish Transfer Unit Methods
12	Methods of Computation
13	2nd Group Meeting
14	Subsystem Revision
15	Memo Report 3
	Preliminary Design Report:
16	Subsystem Screening, Tables, and Descriptions
17	Project Timeline
18	Problem Statement
19	Subsystem Drawings and Descriptions
20	Subsystem Scoring, Tables, and Descriptions
21	Preliminary Calculations and Feasibility Calculations
22	First Hydraulics Lab Tour
23	Design Concepts Selected for Investigation
24	3rd Group Meeting
25	Report Revision
26	Preliminary Design Report Due
	Preliminary Machine Architecture Report:
27	CAD Transfer Unit Subsystem
28	CAD Attachment Type Subsystem
29	Memo Report 4
30	4th Group Meeting
31	Redesign Transfer Unit Subsystem
32	Calculations, Design Requirements, and Final Solution
33	5th Group Meeting

34	Redesign Attachment Subsystem
35	Second Hydraulics Lab Tour
36	6th Group Meeting
37	CAD New Transfer Unit
38	CAD New Attachment Subsystem
39	Compile Tables, CADs, and Assembly Drawings
40	Cost Analysis for Subsystems
41	7th Group Meeting
42	Bill of Materials Due
43	Report Revisions
44	Send Draft to Sponsors for Fish Safe Products
45	Bill of Materials Revision
46	Preliminary Machine Architecture Report Due
	Spring Quarter
	Manufacturing and Testing:
47	Send BOM to Sponsor and Request Material Purchases
48	Materials Purchased
49	Computation Testing
50	Testing Plan
51	Manufacturing and Assembly of Transfer Unit and Attachment Subsystems
52	Refined Testing Plan
53	Computation Re-Training and Validation
54	Testing of Transfer Unit, Sensor, and Computation Subsystems
55	Computation Training from In-Person Testing
56	System Testing

57	Computation Training from In-Person Training
58	Material Arrival
59	Sponsor Update Meeting
60	Hydraulics Lab Meeting
61	Purchases for Missing/Incorrect Materials
62	Poster Designing
63	Poster Due
64	Prepare for Critical Design Reviews
65	Engineering Design Showcase
66	Final Poster, Report, and Critical Design Review

Gantt Chart

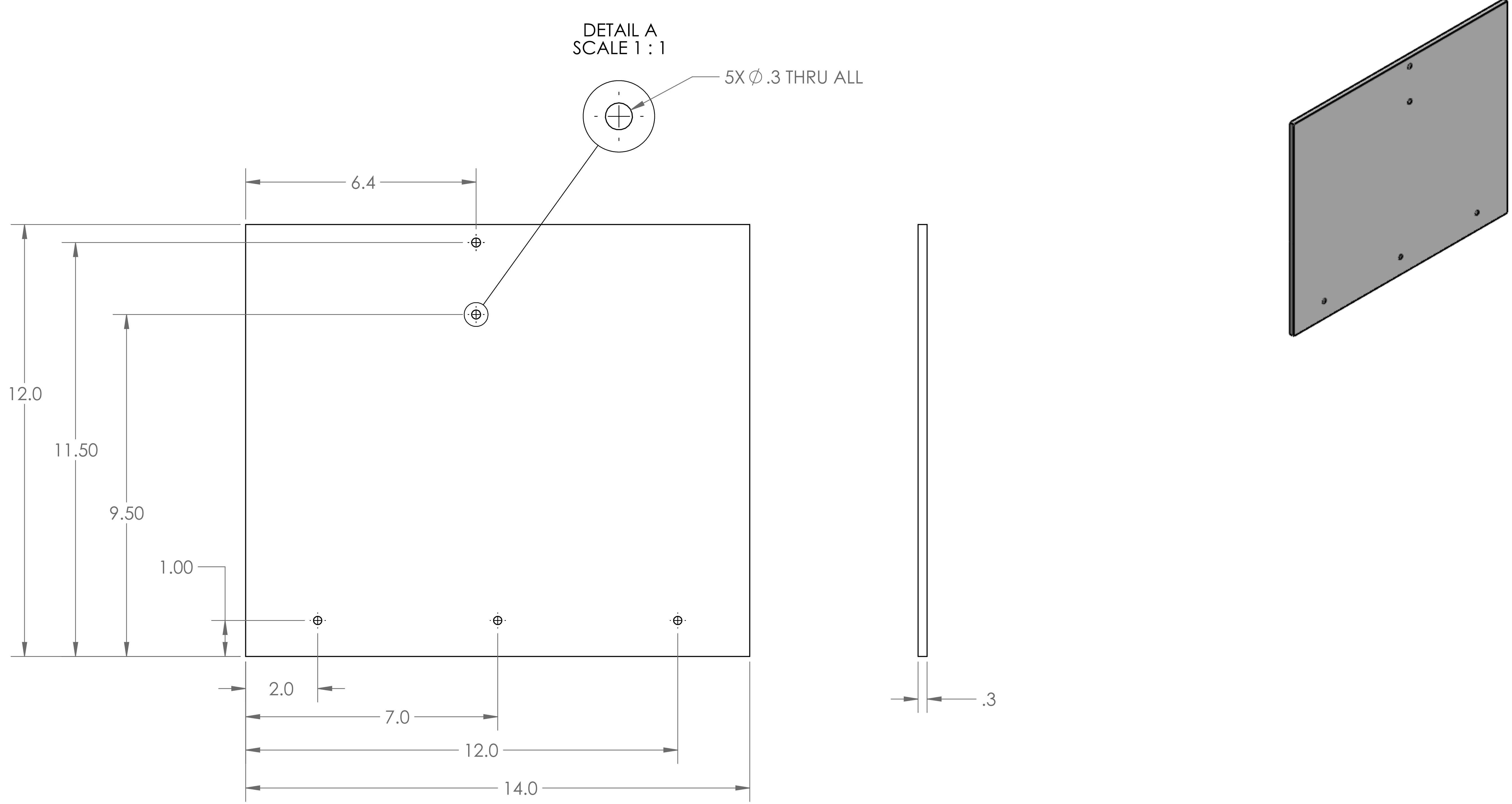
Counting Little Fishies for Science

Read-only view, generated on 06 Jun 2026

Instagantt

	ACTIVITIES	ASSIGNEE	EH	START	DUE	STATUS	%
	Understanding the Problem:		-	**12/Jan**	**30/Jan**		**100%**
1	Prepare Questions for Spon...	Unassigned	-	12/Jan	15/Jan	Finished	100%
2	Memo Report 1	Unassigned	-	12/Jan	14/Jan	Finished	100%
3	1st Sponsor Meeting	Unassigned	-	16/Jan	16/Jan	Finished	100%
4	Sensor Research	Unassigned	-	17/Jan	23/Jan	Finished	100%
5	Brainstorm Solutions	Unassigned	-	17/Jan	23/Jan	Finished	100%
6	Find Solutions already being...	Unassigned	-	17/Jan	30/Jan	Finished	100%
7	1st Group Meeting	Unassigned	-	24/Jan	24/Jan	Finished	100%
8	Memo Report 2	Unassigned	-	25/Jan	26/Jan	Finished	100%
	Subsystem Identification:		-	**25/Jan**	**02/Feb**		**100%**
10	Sensors on the Market	Unassigned	-	25/Jan	30/Jan	Finished	100%
11	Systems Currently in Use	Unassigned	-	25/Jan	30/Jan	Finished	100%
12	Fish Transfer Unit Methods	Unassigned	-	25/Jan	30/Jan	Finished	100%
13	Methods of Computation	Unassigned	-	25/Jan	30/Jan	Finished	100%
14	2nd Group Meeting	Unassigned	-	31/Jan	31/Jan	Finished	100%
15	Subsystem Revision	Unassigned	-	31/Jan	02/Feb	Finished	100%
16	Memo Report 3	Unassigned	-	31/Jan	02/Feb	Finished	100%
	Preliminary Design Report:		-	**31/Jan**	**13/Feb**		**100%**
18	Subsystem Screening, Tabl...	Unassigned	-	31/Jan	02/Feb	Finished	100%
19	Project Timeline	Unassigned	-	31/Jan	06/Feb	Finished	100%
20	Problem Statement	Unassigned	-	01/Feb	07/Feb	Finished	100%
21	Subsystem Drawings and D...	Unassigned	-	02/Feb	09/Feb	Finished	100%
22	Subsystem Scoring, Tables, ...	Unassigned	-	03/Feb	09/Feb	Finished	100%
23	Preliminary Calculations an...	Unassigned	-	05/Feb	11/Feb	Finished	100%
24	First Hydraulics Lab Tour	Unassigned	-	05/Feb	05/Feb	Finished	100%
25	Design Concepts Selected f...	Unassigned	-	06/Feb	09/Feb	Finished	100%
26	3rd Group Meeting	Unassigned	-	07/Feb	07/Feb	Finished	100%
27	Report Revision	Unassigned	-	11/Feb	13/Feb	Finished	100%
28	Preliminary Design Report ...	Unassigned	-	13/Feb	13/Feb	Finished	100%
	Preliminary Machine Architectu...		-	**14/Feb**	**20/Mar**		**100%**
30	CAD Transfer Unit Subsyste...	Unassigned	-	14/Feb	20/Feb	Finished	100%
31	CAD Attachment Type Subs...	Unassigned	-	14/Feb	20/Feb	Finished	100%
32	Memo Report 4	Unassigned	-	21/Feb	02/Mar	Finished	100%
33	4th Group Meeting	Unassigned	-	21/Feb	21/Feb	Finished	100%
34	Redesign Transfer Unit Sub...	Unassigned	-	21/Feb	11/Mar	Finished	100%
35	Calculations, Design Requir...	Unassigned	-	21/Feb	20/Mar	Finished	100%
36	5th Group Meeting	Unassigned	-	28/Feb	28/Feb	Finished	100%
37	Redesign Attachment Subsy...	Unassigned	-	28/Feb	11/Mar	Finished	100%
38	Second Hydraulics Lab Tour	Unassigned	-	05/Mar	05/Mar	Finished	100%
39	6th Group Meeting	Unassigned	-	07/Mar	07/Mar	Finished	100%
40	CAD New Transfer Unit	Unassigned	-	08/Mar	16/Mar	Finished	100%
41	CAD New Attachment Subs...	Unassigned	-	08/Mar	16/Mar	Finished	100%
42	Compile Tables, CADs, and ...	Unassigned	-	12/Mar	16/Mar	Finished	100%
43	Cost Analysis for Subsystems	Unassigned	-	13/Mar	16/Mar	Finished	100%
44	7th Group Meeting	Unassigned	-	14/Mar	14/Mar	Finished	100%
45	Bill of Materials Revision	Unassigned	-	16/Mar	20/Mar	Finished	100%
46	Bill of Materials Due	Unassigned	-	16/Mar	16/Mar	Finished	100%
47	Report Revisions	Unassigned	-	16/Mar	20/Mar	Finished	100%
48	Preliminary Machine Archit...	Unassigned	-	20/Mar	20/Mar	Finished	100%
	Manufacturing and Testing:		-	**24/Mar**	**05/Jun**		**100%**
50	Send BOM to Sponsor and R...	Unassigned	-	24/Mar	01/Apr	Finished	100%
51	Materials Purchased	Unassigned	-	02/Apr	02/Apr	Finished	100%
	Assembly and Integration of...	Unassigned	-	03/Apr	04/Jun	Finished	100%
53	Computation Testing	Unassigned	-	03/Apr	10/May	Finished	100%
54	Testing Plan	Unassigned	-	15/Apr	29/Apr	Finished	100%
55	Manufacturing and Ass...	Unassigned	-	24/Apr	21/May	Finished	100%
56	Refined Testing Plan	Unassigned	-	30/Apr	04/May	Finished	100%
57	Computation Re-Trainin...	Unassigned	-	11/May	21/May	Finished	100%
58	Testing of Transfer Unit...	Unassigned	-	22/May	22/May	Finished	100%
59	Computation Training fr...	Unassigned	-	23/May	28/May	Finished	100%
60	System Testing	Unassigned	-	29/May	29/May	Finished	100%
61	Computation Training fr...	Unassigned	-	30/May	04/Jun	Finished	100%
62	Material Arrival	Unassigned	-	06/Apr	15/Apr	Finished	100%
63	Sponsor Update Meeting	Unassigned	-	17/Apr	17/Apr	Finished	100%
64	Hydraulics Lab Meeting	Unassigned	-	20/Apr	20/Apr	Finished	100%
65	Purchases for Missing/Incor...	Unassigned	-	21/Apr	21/Apr	Finished	100%
66	Poster Designing	Unassigned	-	11/May	15/May	Finished	100%
67	Poster Due	Unassigned	-	16/May	18/May	Finished	100%
68	Prepare for Critical Design ...	Unassigned	-	25/May	03/Jun	Finished	100%
69	Engineering Design Showca...	Unassigned	-	04/Jun	04/Jun	Finished	100%
70	Final Poster, Report, and Cri...	Unassigned	-	05/Jun	05/Jun	Finished	100%

Appendix III: Engineering Drawings of Systems of Parts Manufactured



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:

LINEAR	ANGULAR
X.X = ±0.1	X = ±1°
X.XX = ±0.03	X.X = ±0.5°
X.XXX = ±0.005	

	NAME	DATE
DESIGNED	HN	3/12/2026
DRAWN	NK	3/16/2026
MFG APPR.		

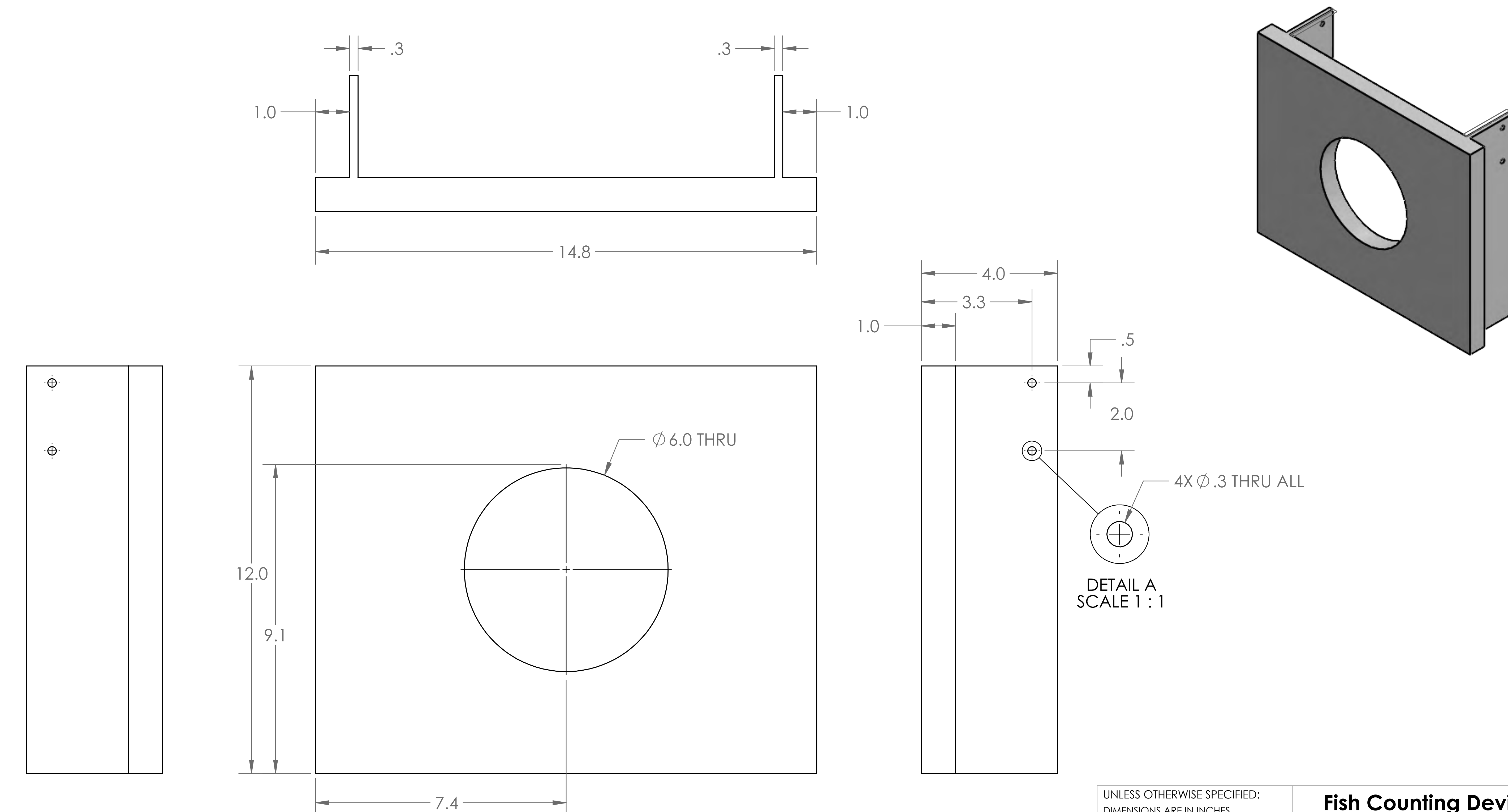
Fish Counting Device

TITLE:

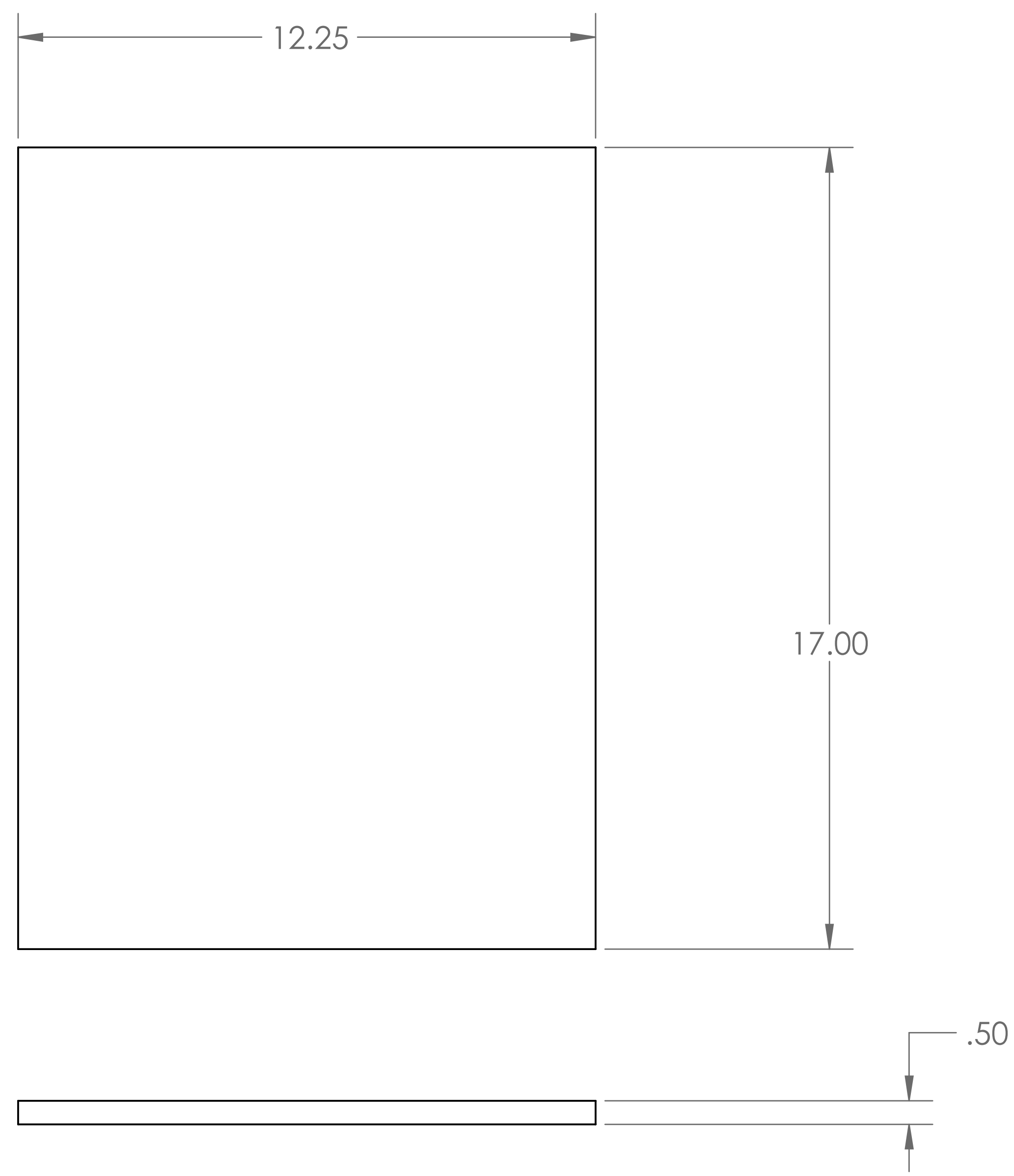
Side Wall(TU)

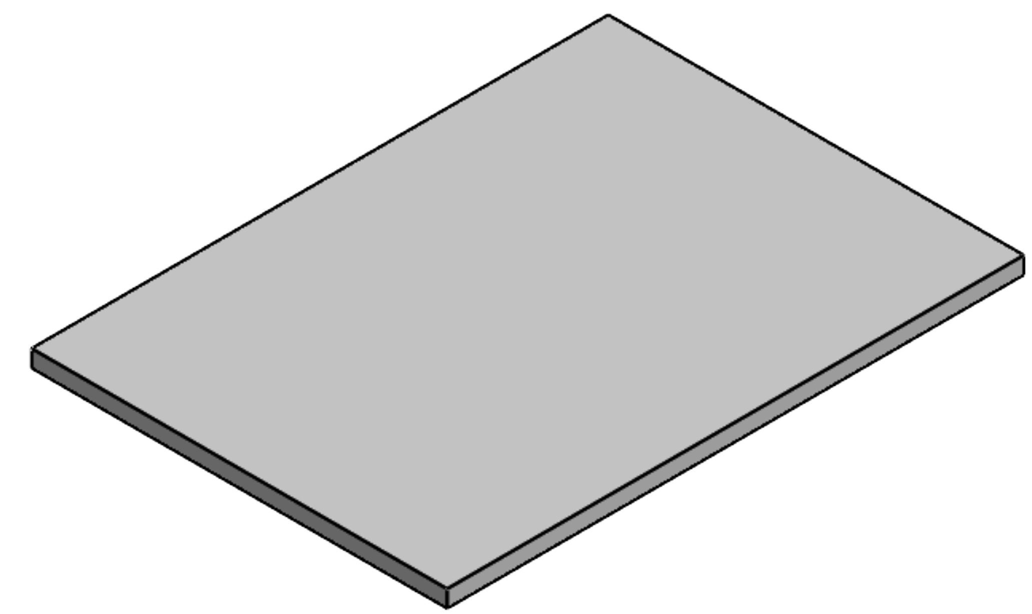
SIZE **B** MATERIAL PVC Rigid REV 1

SCALE: 1:3 SHEET 1 OF 1



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: LINEAR X.X = ±0.1 X.XX = ±0.03 X.XXX = ±0.005 ANGULAR X = ±1° X.X = ±0.5°			**Fish Counting Device**
			TITLE: Entry(TU)
	NAME	DATE	
DESIGNED	HN	3/09/2026	SIZE **B** MATERIAL PVC Rigid REV 1
DRAWN	NK	3/16/2026	
MFG APPR.			SCALE: 1:3 SHEET 1 OF 1





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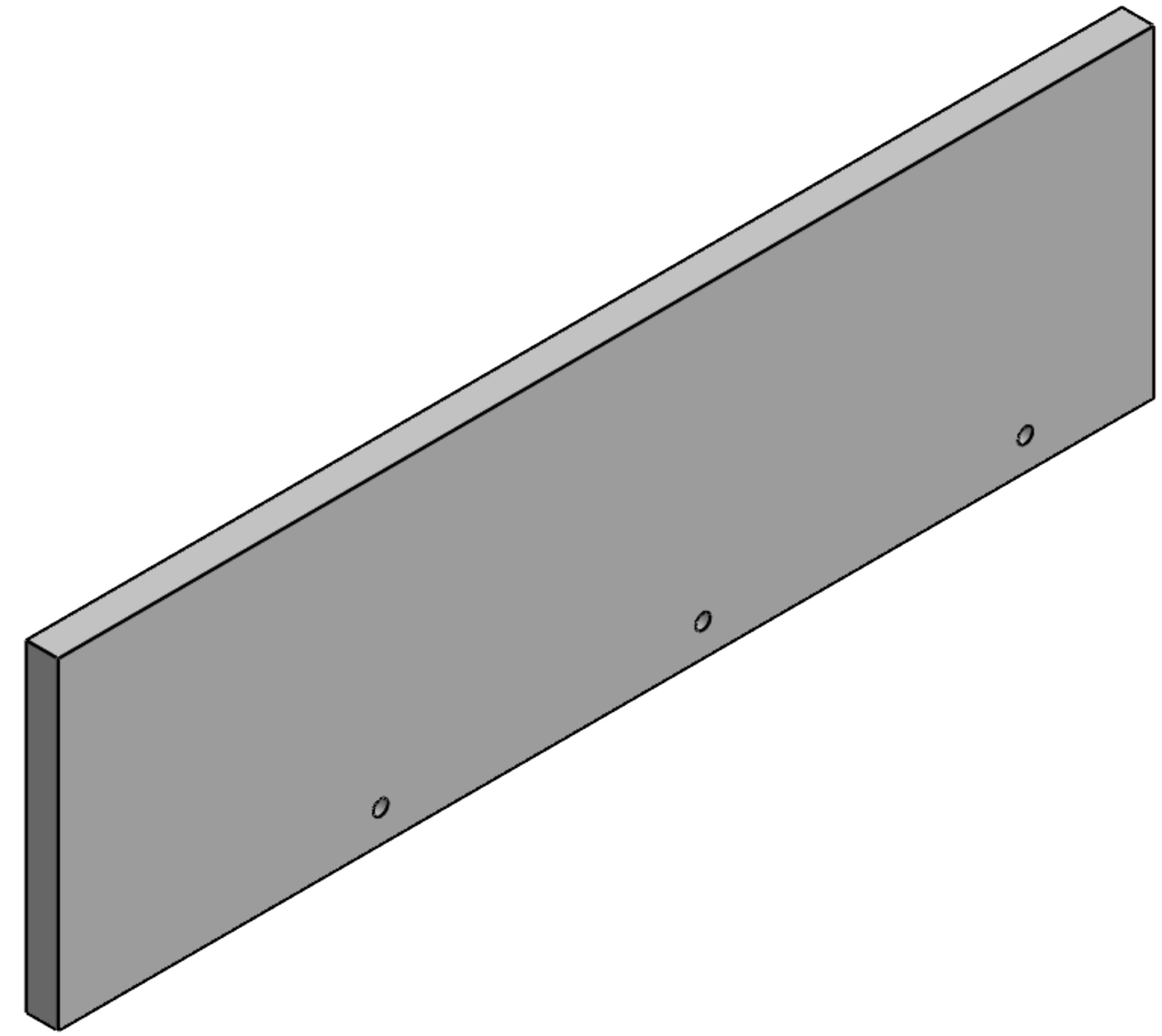
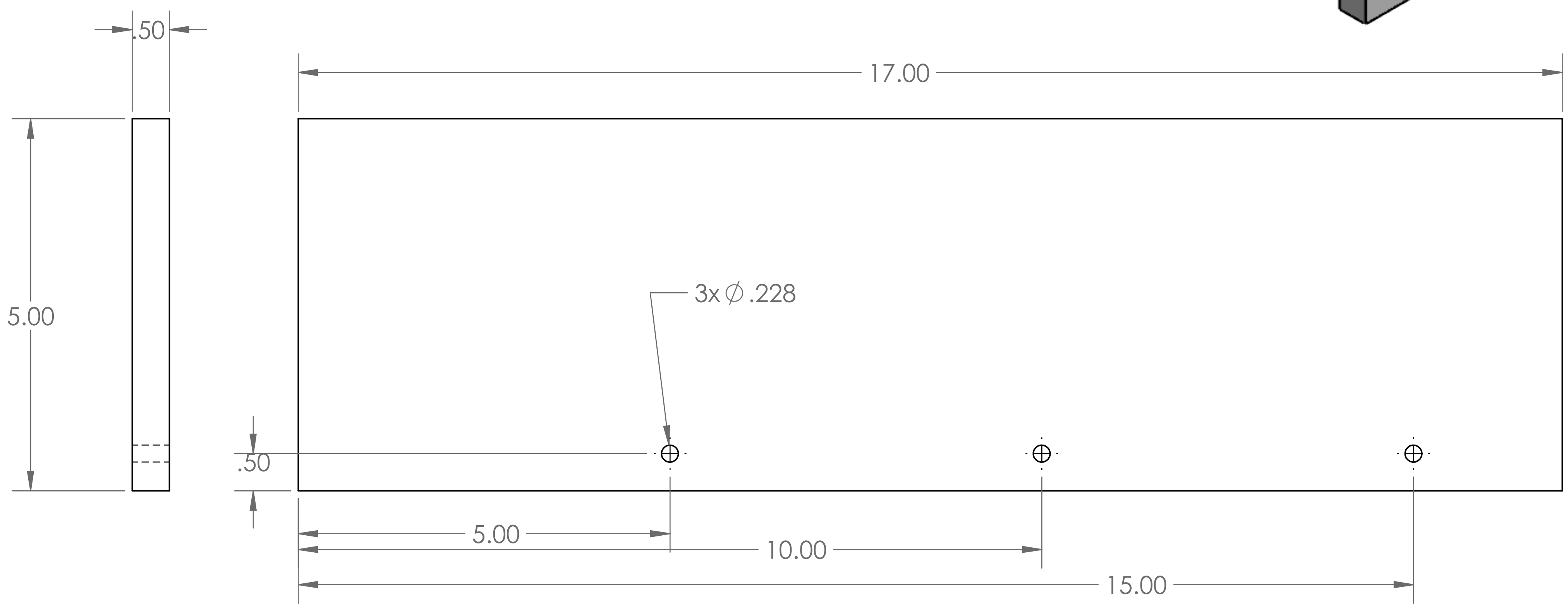
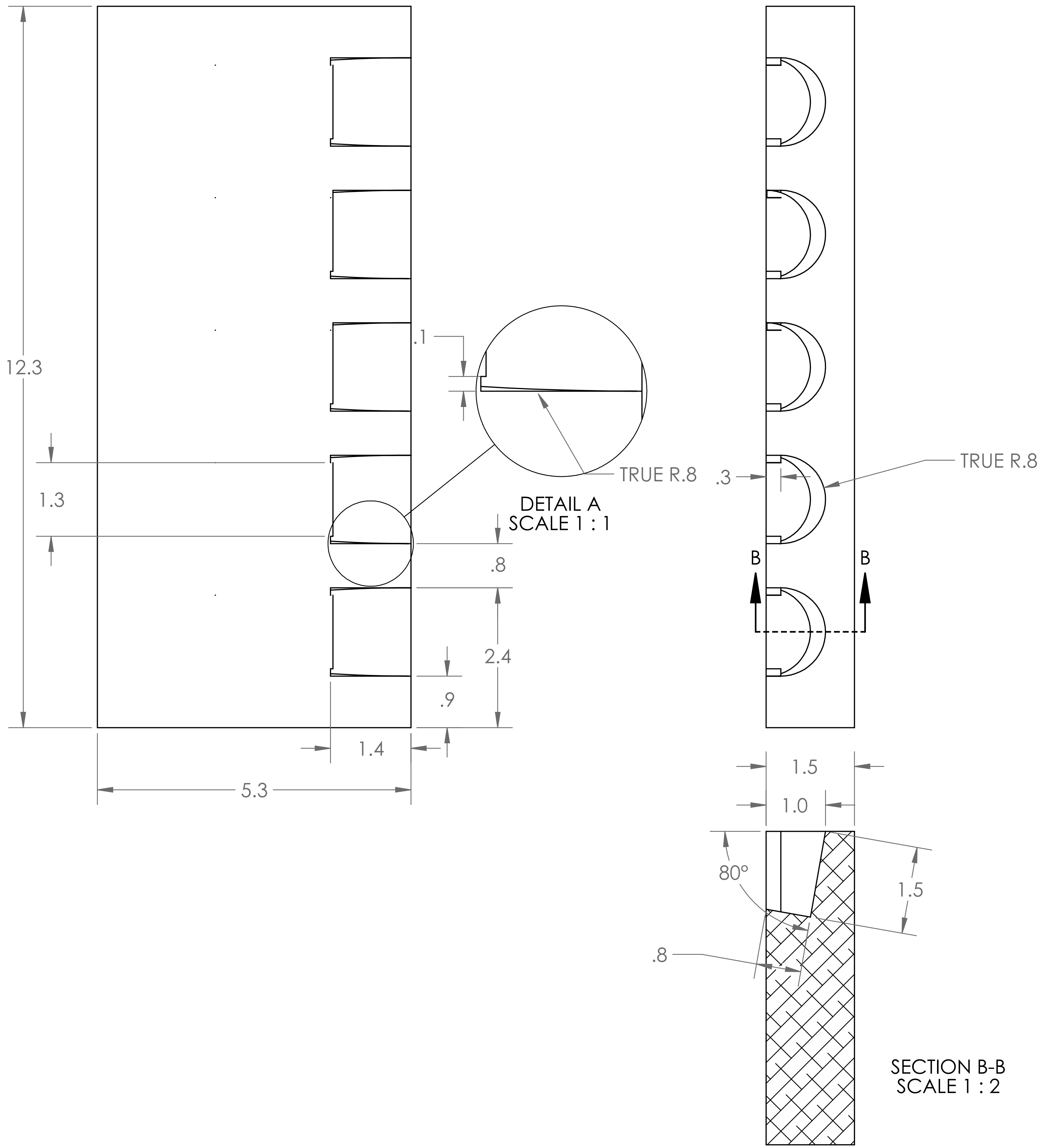
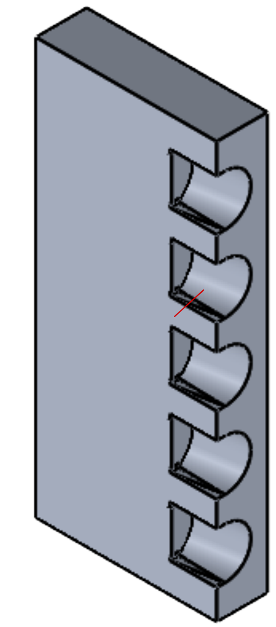
	NAME	DATE	TITLE: Base Plate
DESIGNED	CLV	3/16/2026	SIZE **B** · MATERIAL PVC Rigid · REV 1
DRAWN	CLV	3/16/2026	
MFG APPR.			SCALE: 1:4 · SHEET 1 OF 1





UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MM TOLERANCES: LINEAR X.X = ±0.5 X.XX = ±0.25 X.XXX = ±0.125 ANGULAR X = ±1° X.X = ±0.5°			Fish Counting Device		
	NAME	DATE	TITLE: Base Plate Connection		
DESIGNED	CLV	3/16/2026	SIZE B	MATERIAL PVC Rigid	REV 1
DRAWN	CLV	3/16/2026			
MFG APPR.			SCALE: 1:4	SHEET 1 OF 1	





UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:

LINEAR	ANGULAR
X.X = ±0.1	X = ±1°
X.XX = ±0.03	X.X = ±0.5°
X.XXX = ±0.005	

	NAME	DATE
DESIGNED	CL	3/14/2026
DRAWN	NK	3/16/2026
MFG APPR.		

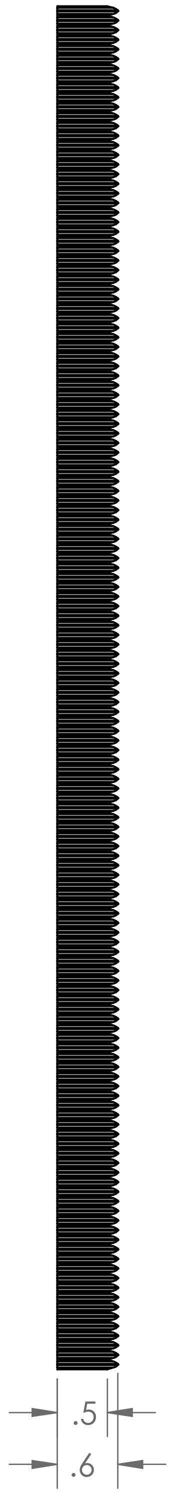
Fish Counting Device

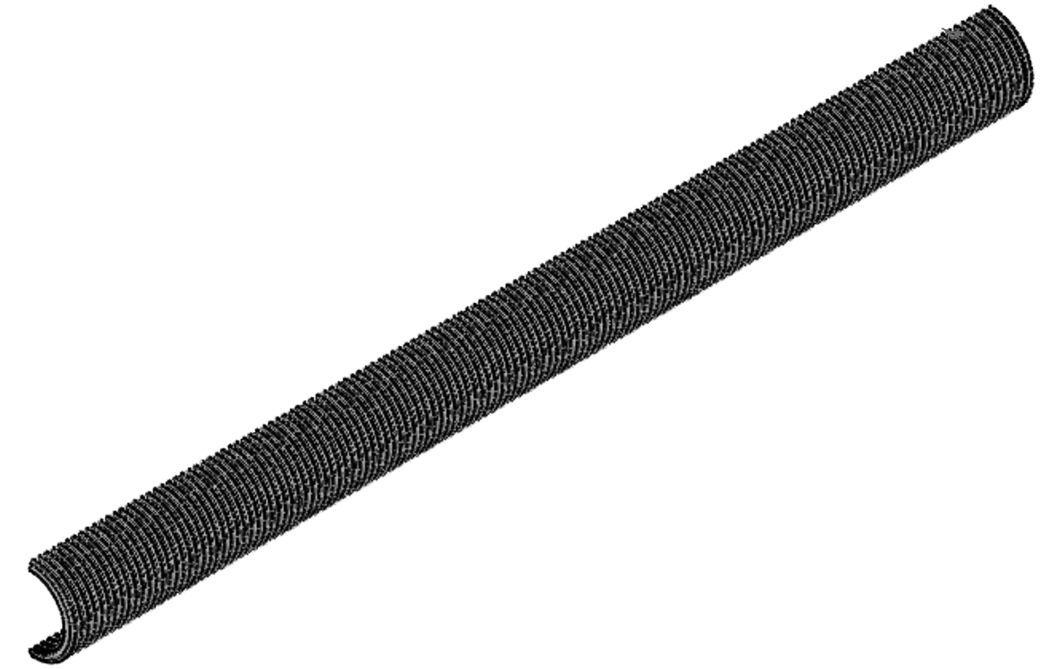
TITLE:

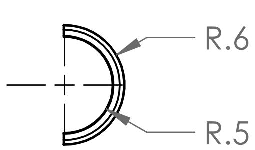
Lane Connector

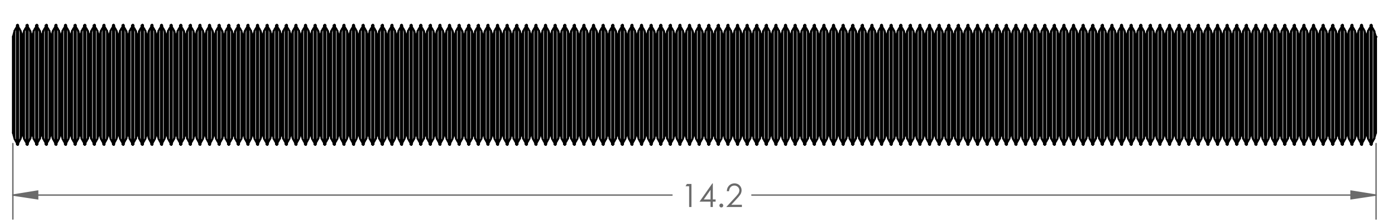
SIZE	MATERIAL	REV
B	Aluminum 6061 Alloy	1

SCALE: 1:5 SHEET 1 OF 1

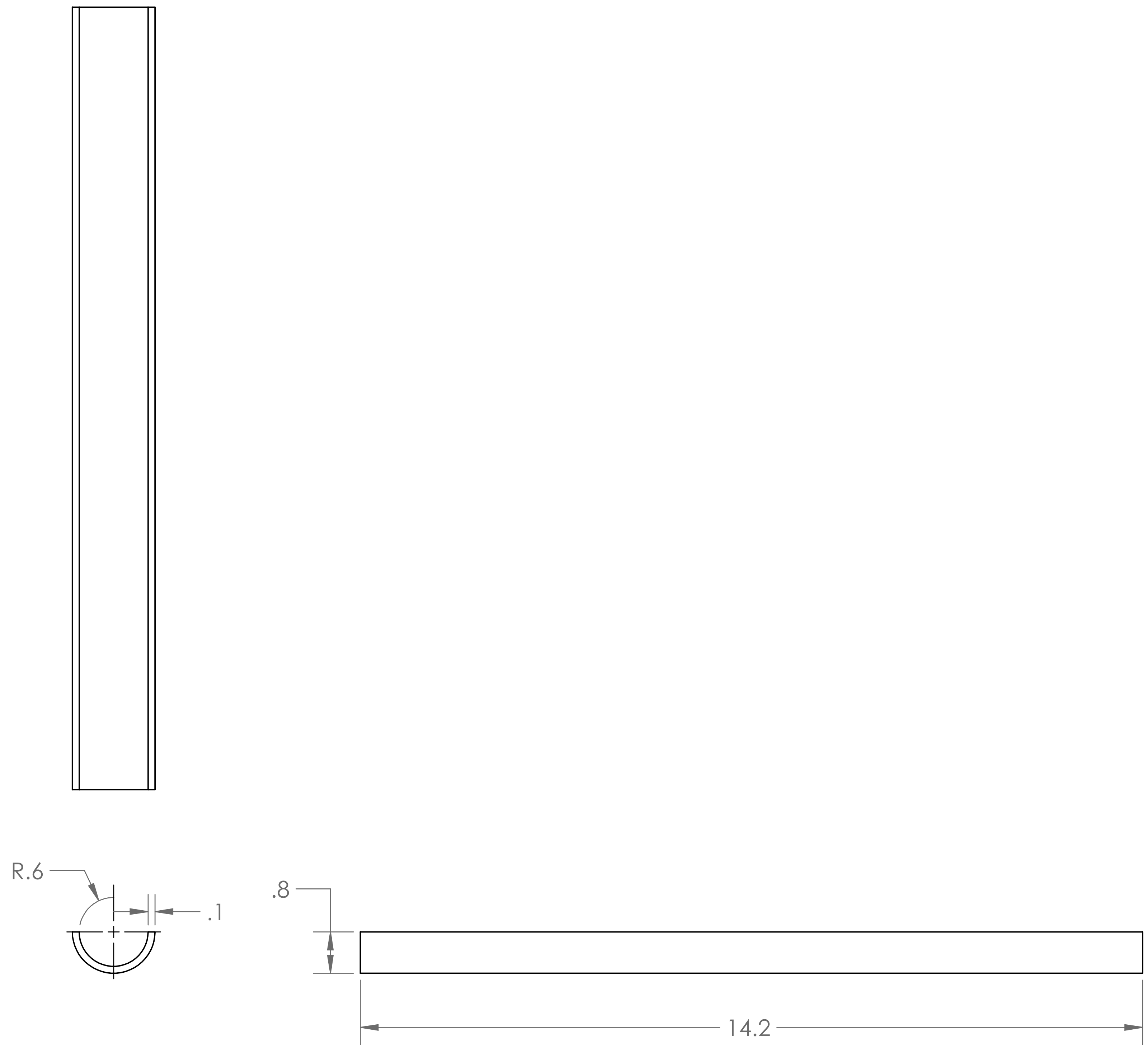


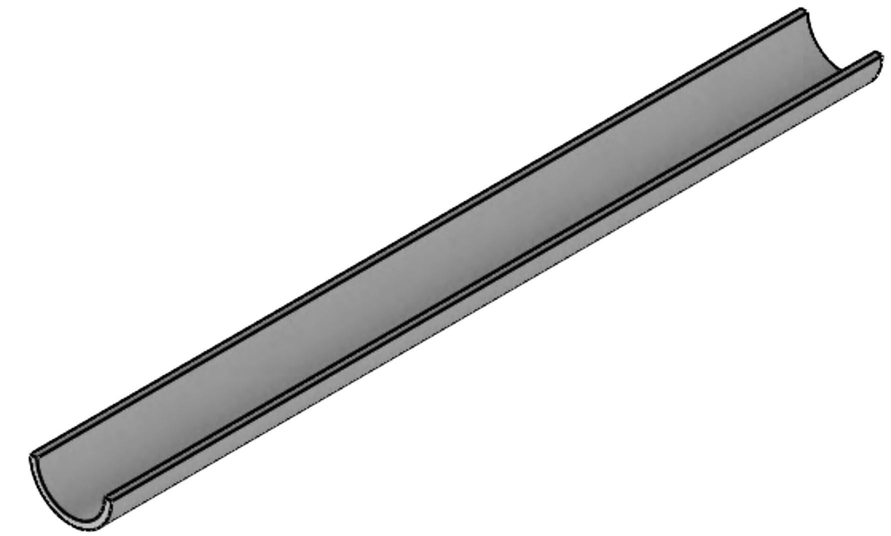




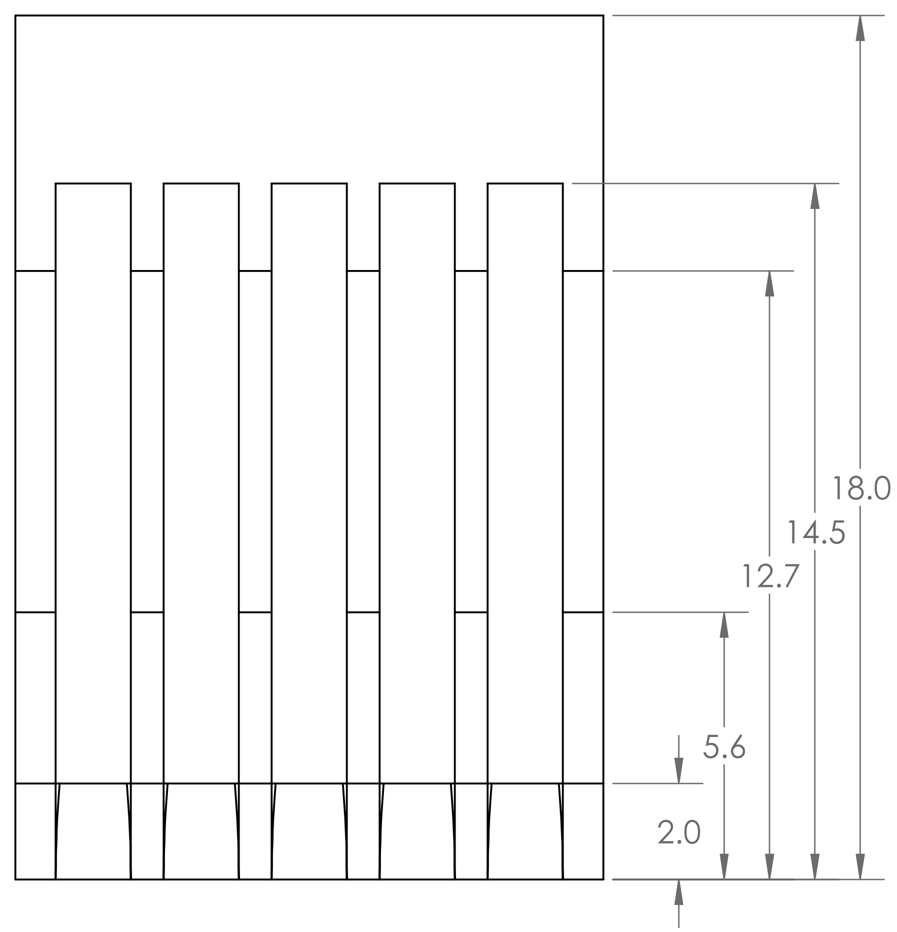


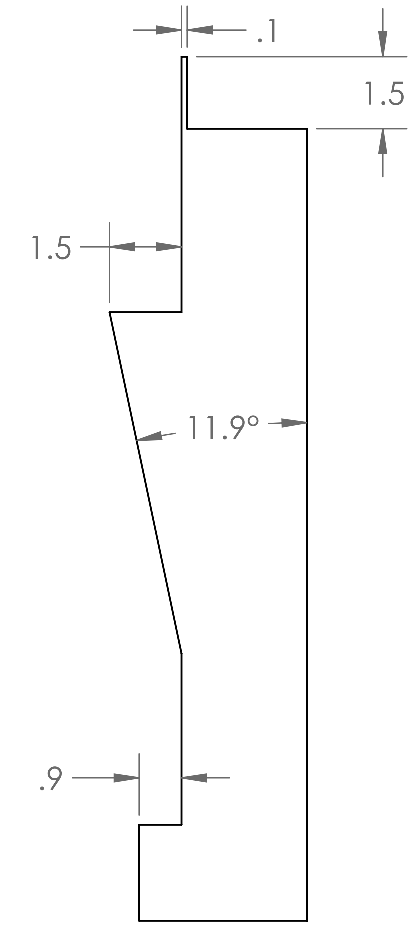
UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: LINEAR X.X = ±0.1 X.XX = ±0.03 X.XXX = ±0.005 ANGULAR X = ±1° X.X = ±0.5°			**Fish Counting Device**
	NAME	DATE	TITLE: Corrugated Pipe
DESIGNED	CL	3/13/2026	SIZE **B** MATERIAL PVC 0.007 Plasticized REV 1
DRAWN	NK	3/16/2026	
MFG APPR.			SCALE: 1:5 SHEET 1 OF 1

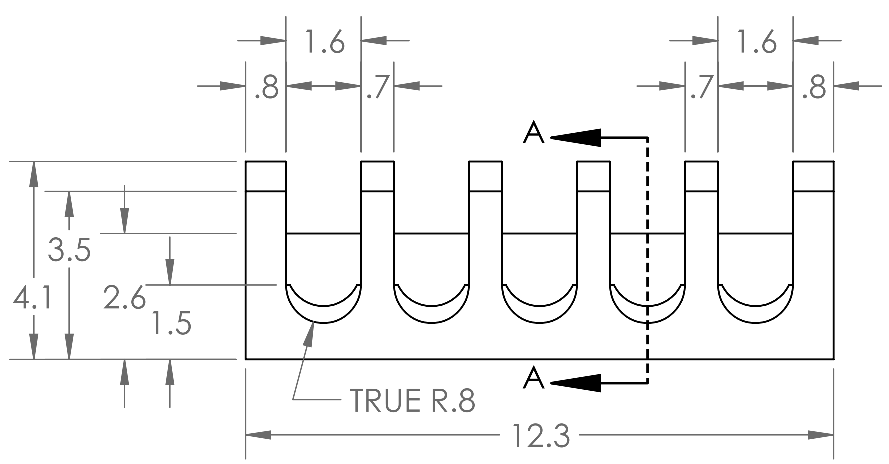


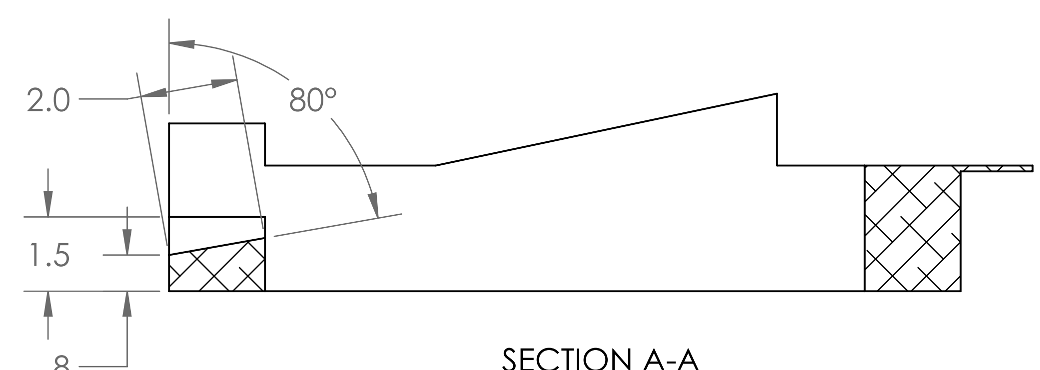


UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: LINEAR X.X = ±0.1 X.XX = ±0.03 X.XXX = ±0.005 ANGULAR X = ±1° X.X = ±0.5°			Fish Counting Device		
	NAME	DATE	TITLE: Lane Holder		
DESIGNED	CL	3/13/2026	SIZE B	MATERIAL PVC Rigid	REV 1
DRAWN	NK	3/16/2026			
MFG APPR.			SCALE: 1:3	SHEET 1 OF 1	

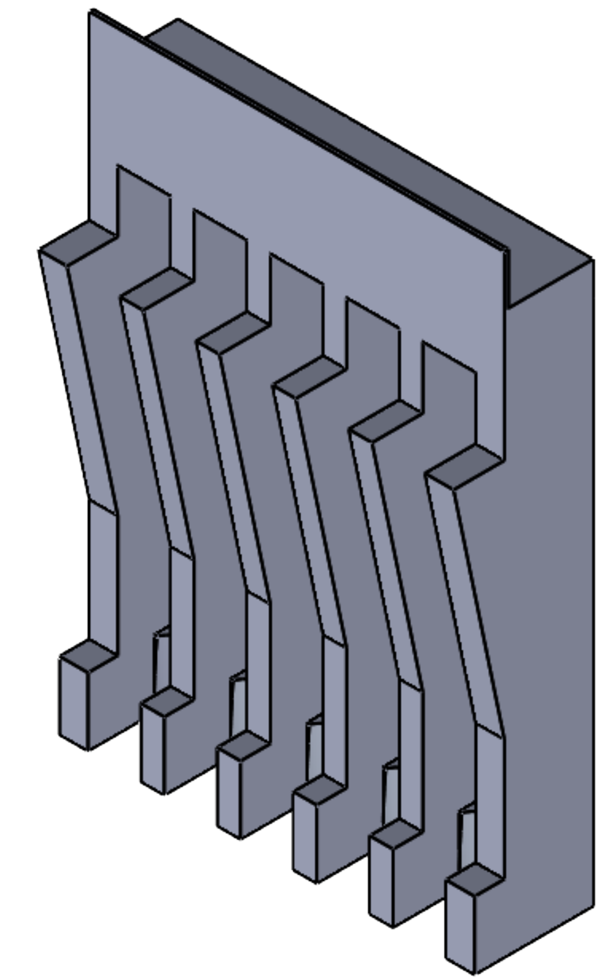








SECTION A-A
SCALE 1 : 4



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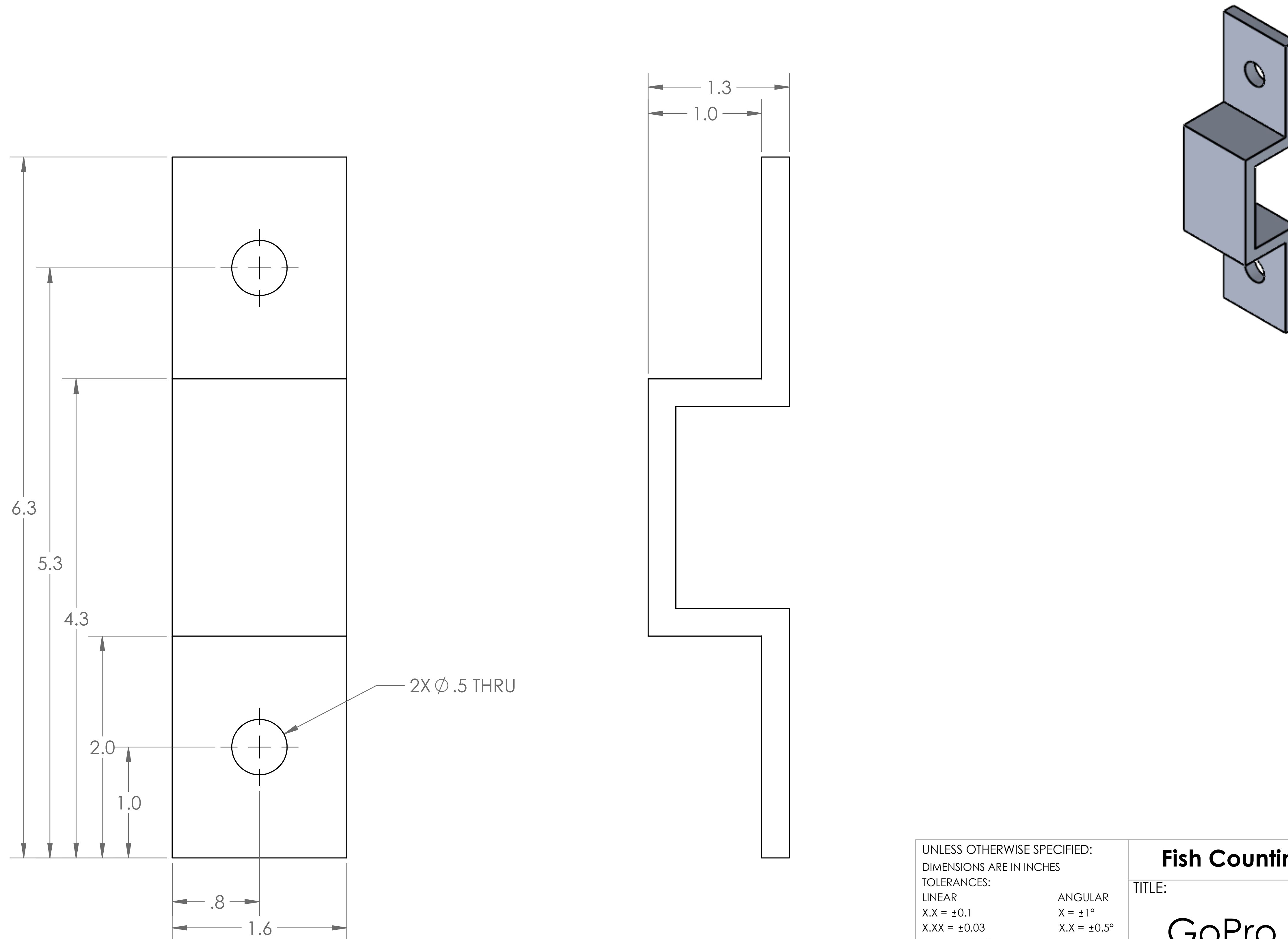
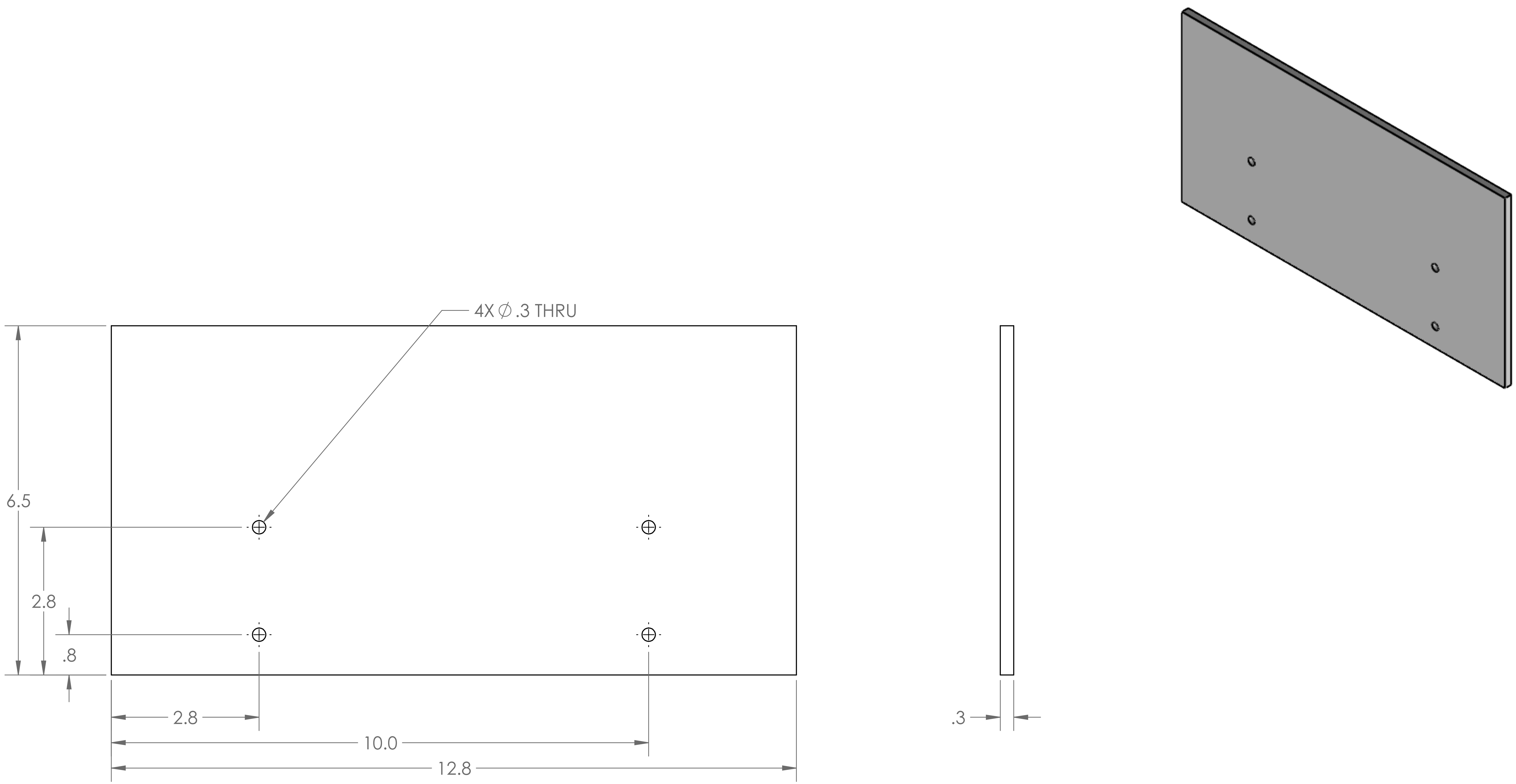
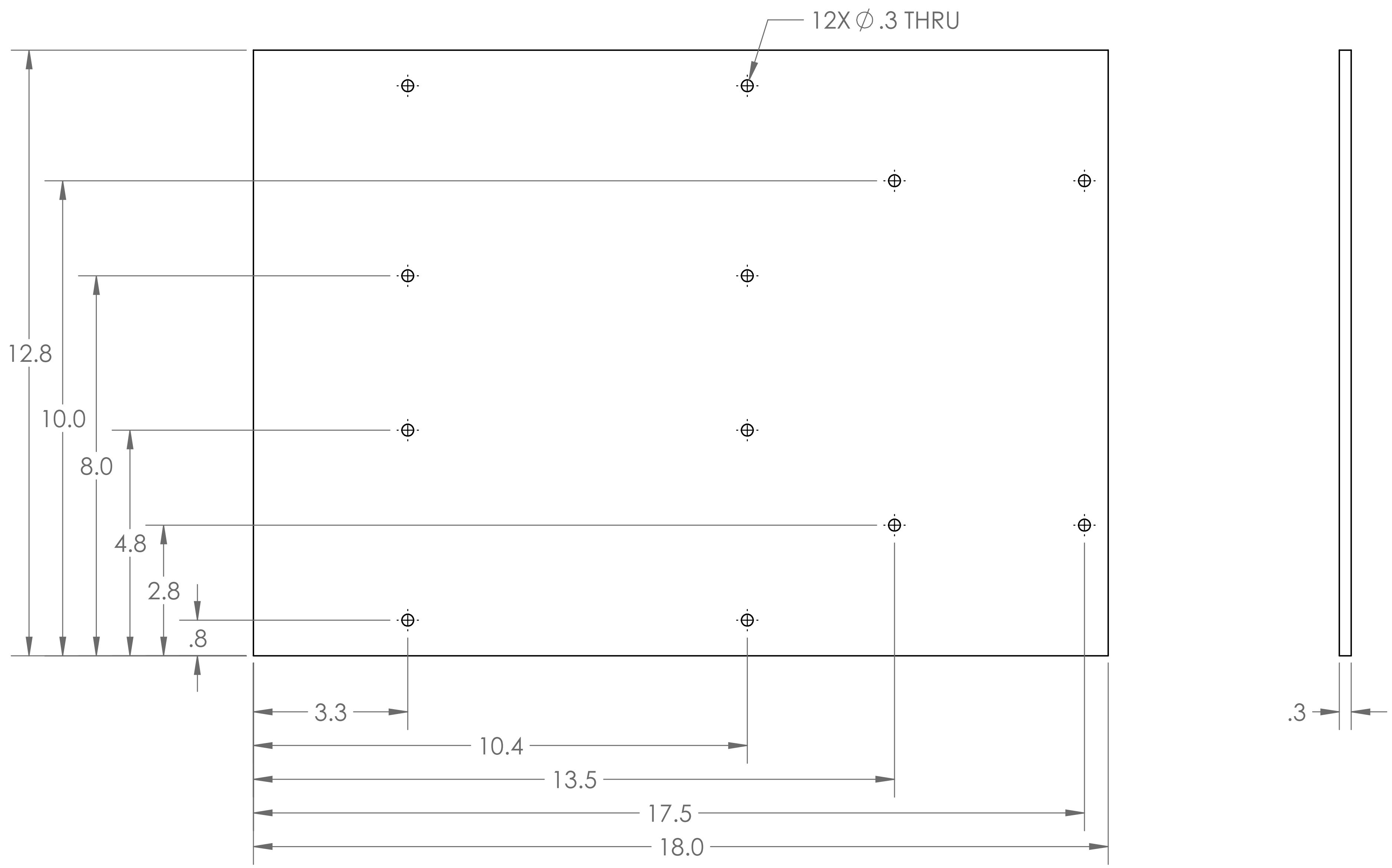
	NAME	DATE	
DESIGNED	CL	3/14/2026	SIZE **B** MATERIAL PLA Plastic REV 1
DRAWN	NK	3/16/2026	
MFG APPR.			SCALE: 1:4 SHEET 1 OF 1

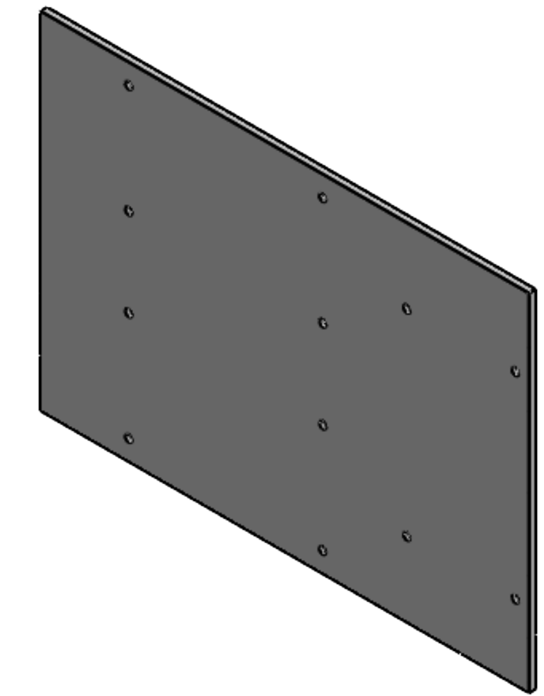


UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: LINEAR X.X = ±0.1 X.XX = ±0.03 X.XXX = ±0.005		ANGULAR X = ±1° X.X = ±0.5°	**Fish Counting Device**	
			TITLE: GoPro Holder	
	NAME	DATE		
DESIGNED	HN	3/12/2026	SIZE **B**	MATERIAL PLA Plastic	REV 1
DRAWN	NK	3/16/2026	SCALE: 1:2	SHEET 1 OF 1
MFG APPR.				

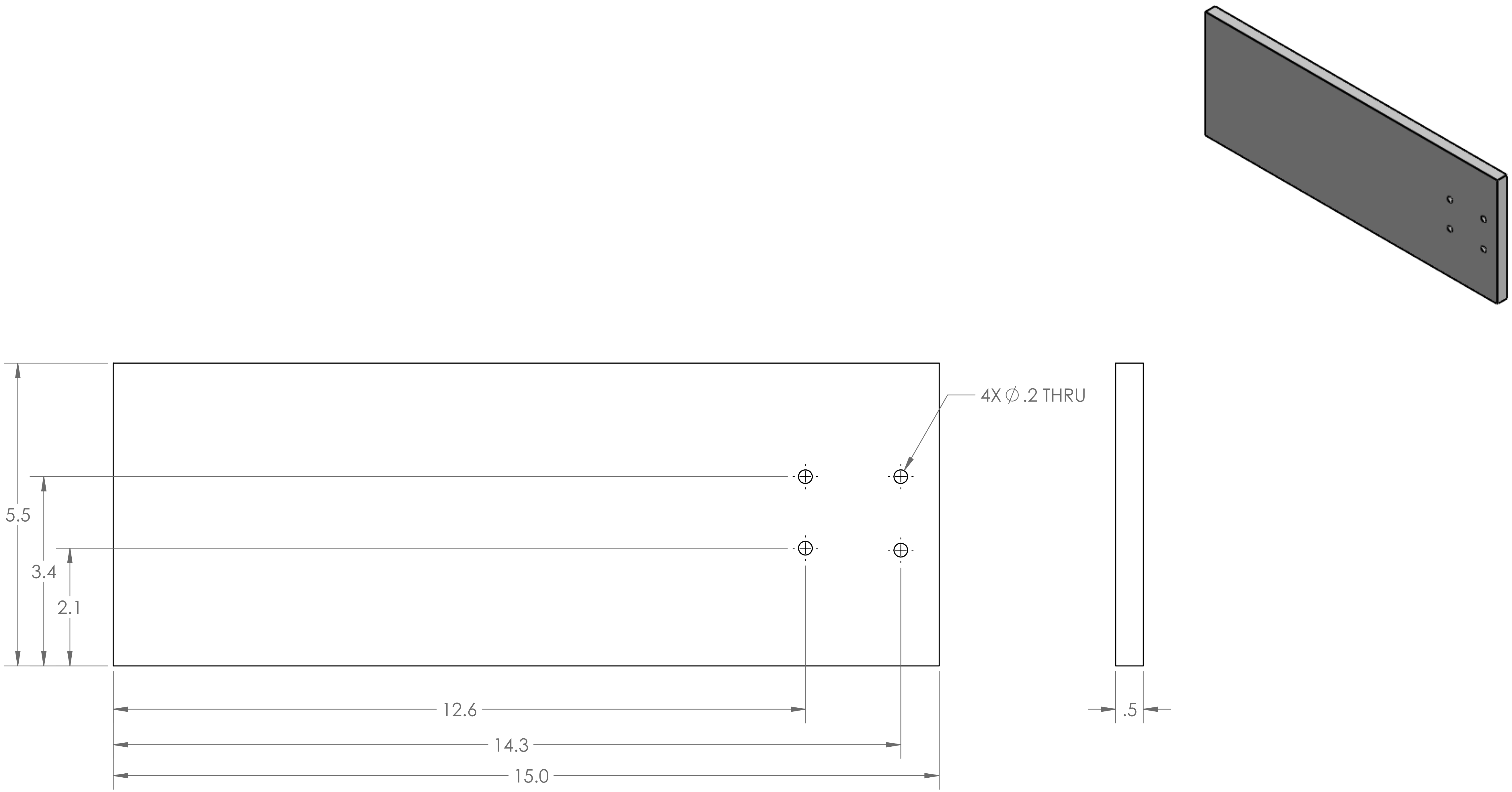


UNLESS OTHERWISE SPECIFIED:			Fish Counting Device		
DIMENSIONS ARE IN INCHES TOLERANCES: LINEAR: X.X = ±0.1, X.XX = ±0.03, X.XXX = ±0.005 ANGULAR: X = ±1°, X.X = ±0.5°			TITLE: Wall Between Systems		
	NAME	DATE	SIZE	MATERIAL	REV
DESIGNED	CL	3/14/2026	B	PVC Rigid	1
DRAWN	NK	3/16/2026			
MFG APPR.			SCALE: 1:2	SHEET 1 OF 1	





UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: LINEAR X.X = ±0.1 X.XX = ±0.03 X.XXX = ±0.005 ANGULAR X = ±1° X.X = ±0.5°			**Fish Counting Device**
			TITLE: Top Roof
	NAME	DATE	SIZE **B** MATERIAL PVC Rigid REV 1
DESIGNED	CL	3/14/2026	
DRAWN	NK	3/16/2026	
MFG APPR.			SCALE: 1:3 SHEET 1 OF 1



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:

LINEAR	ANGULAR
X.X = ±0.1	X = ±1°
X.XX = ±0.03	X.X = ±0.5°
X.XXX = ±0.005	

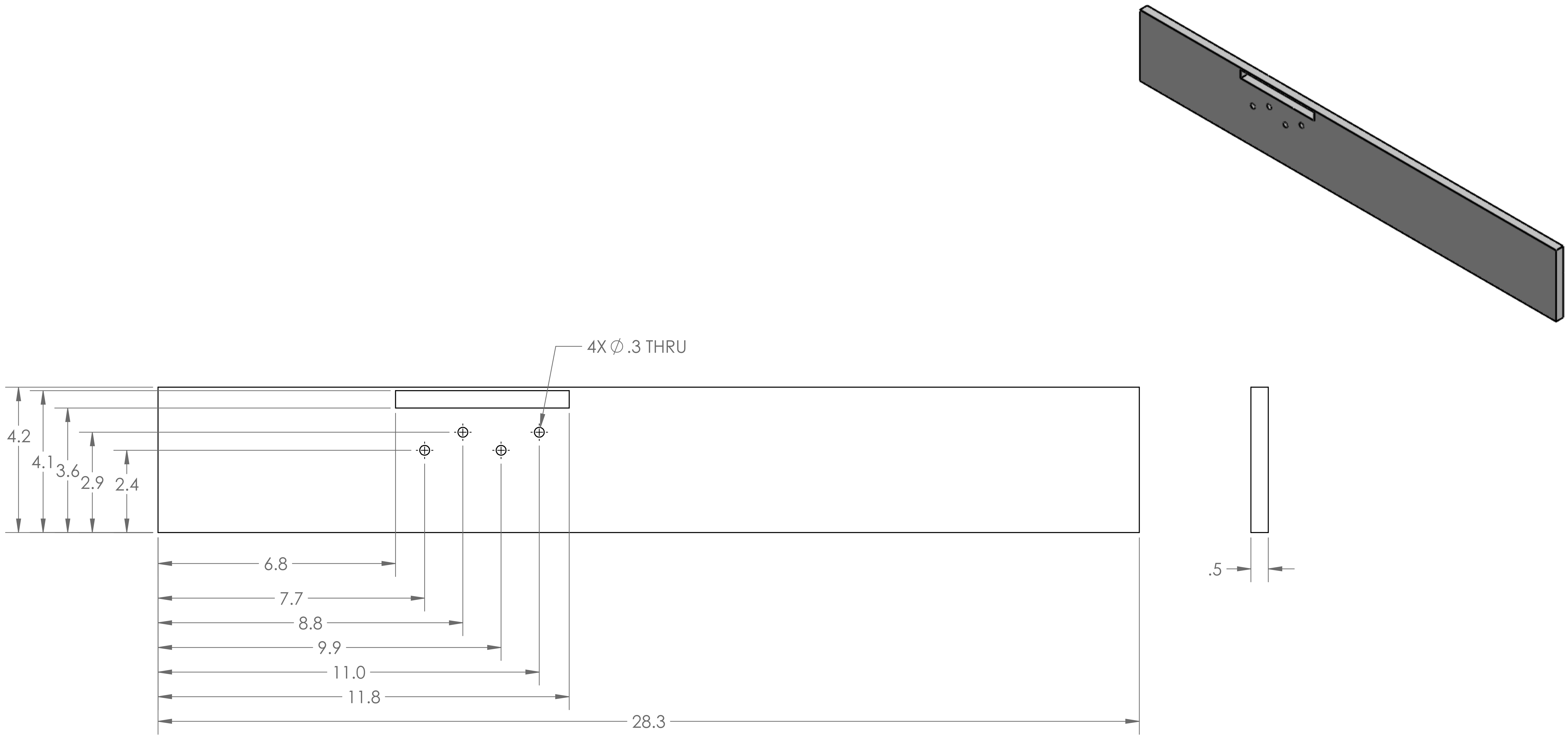
	NAME	DATE
DESIGNED	CL	3/14/2026
DRAWN	NK	3/16/2026
MFG APPR.		

Fish Counting Device

TITLE: Random Plate for Hinge With No Holes for Bracer

SIZE	MATERIAL	REV
B	PVC Rigid	1

SCALE: 1:2 SHEET 1 OF 1



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:

LINEAR	ANGULAR
X.X = ±0.1	X = ±1°
X.XX = ±0.03	X.X = ±0.5°
X.XXX = ±0.005	

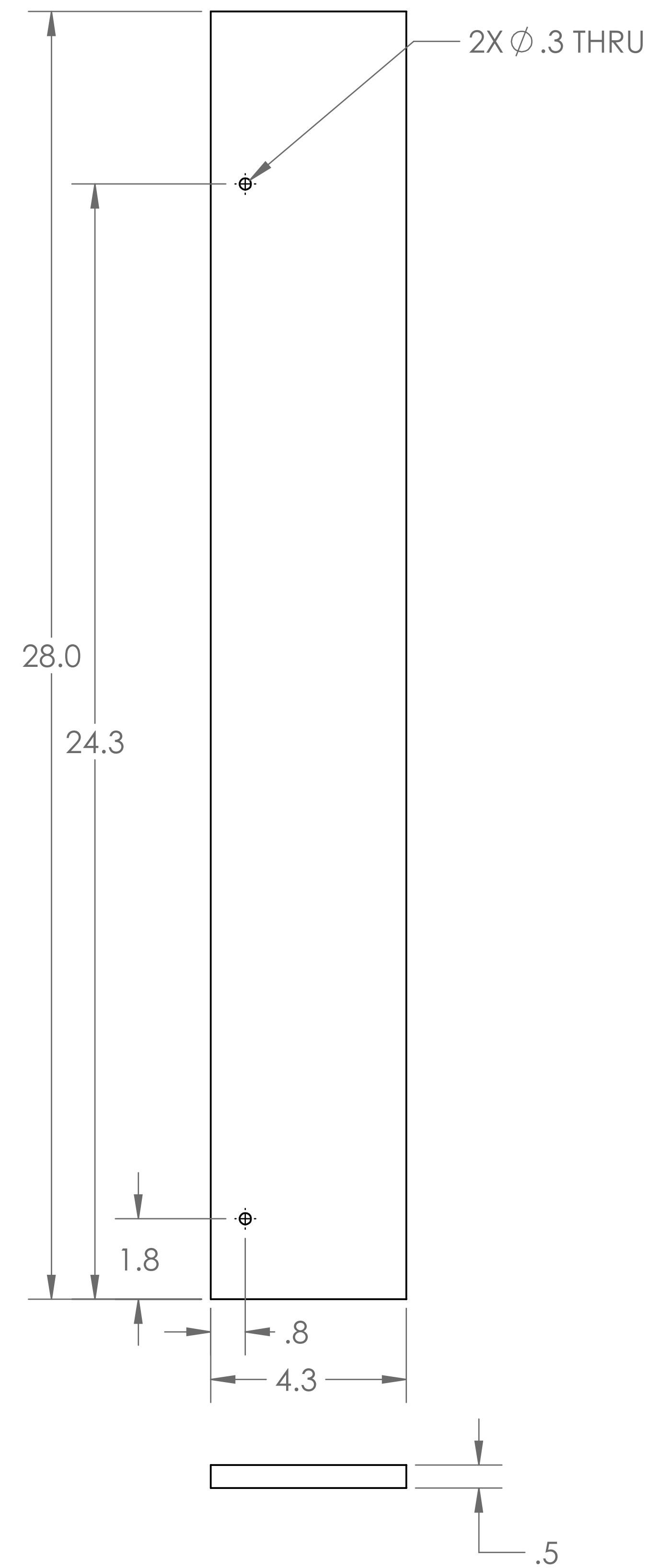
	NAME	DATE
DESIGNED	CL	3/14/2026
DRAWN	NK	3/16/2026
MFG APPR.		

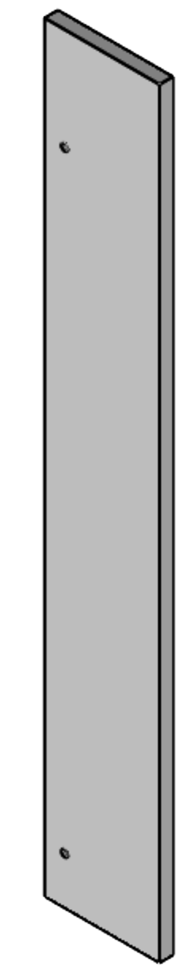
Fish Counting Device

TITLE:
Random Plate for Hinge

SIZE	MATERIAL	REV
B	PVC Rigid	1

SCALE: 1:3 SHEET 1 OF 1





UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:
LINEAR
X.X = ±0.1
X.XX = ±0.03
X.XXX = ±0.005
ANGULAR
X = ±1°
X.X = ±0.5°

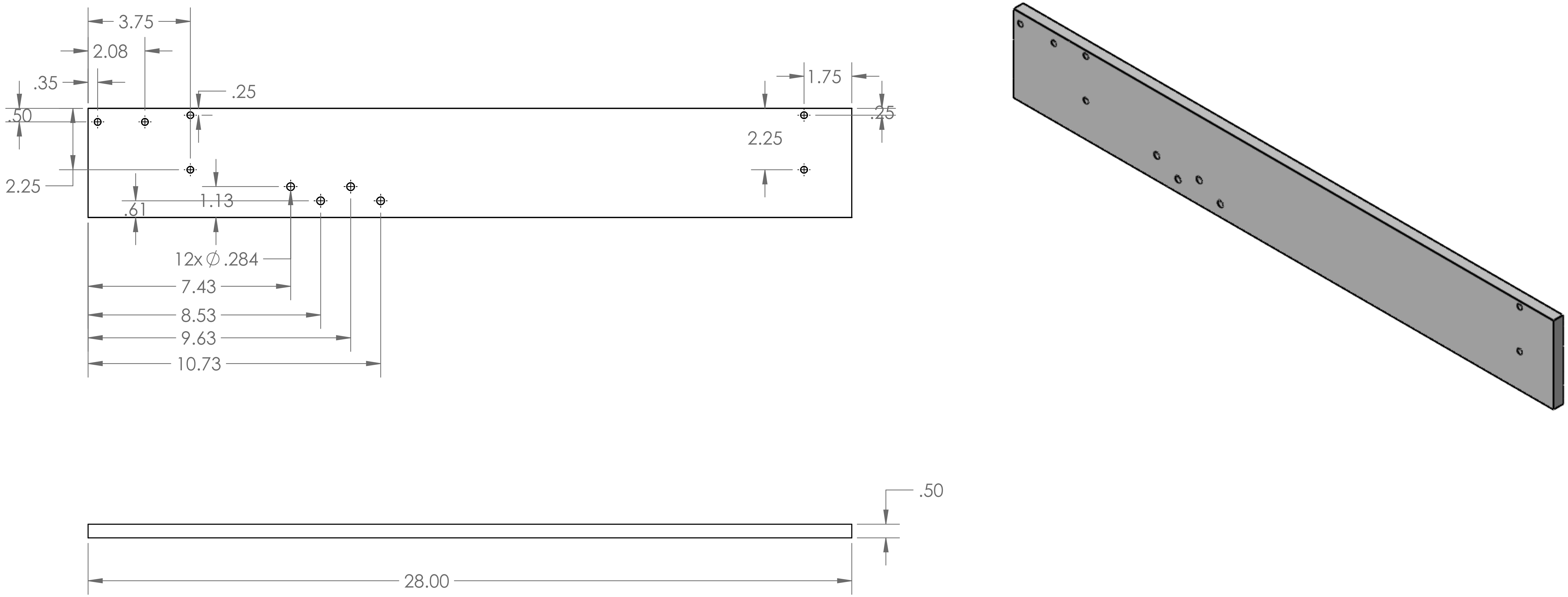
	NAME	DATE
DESIGNED	HN	3/11/2026
DRAWN	NK	3/16/2026
MFG APPR.		

Fish Counting Device

TITLE:
Roof 0.4 Horizontal (TU)

SIZE **B**
MATERIAL PVC Rigid
REV 1

SCALE: 1:4
SHEET 1 OF 1



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MM
TOLERANCES:

LINEAR	ANGULAR
X.X = ±0.5	X = ±1°
X.XX = ±0.25	X.X = ±0.5°
X.XXX = ±0.125	

	NAME	DATE
DESIGNED	CLV	3/14/2026
DRAWN	CLV	3/16/2026
MFG APPR.		

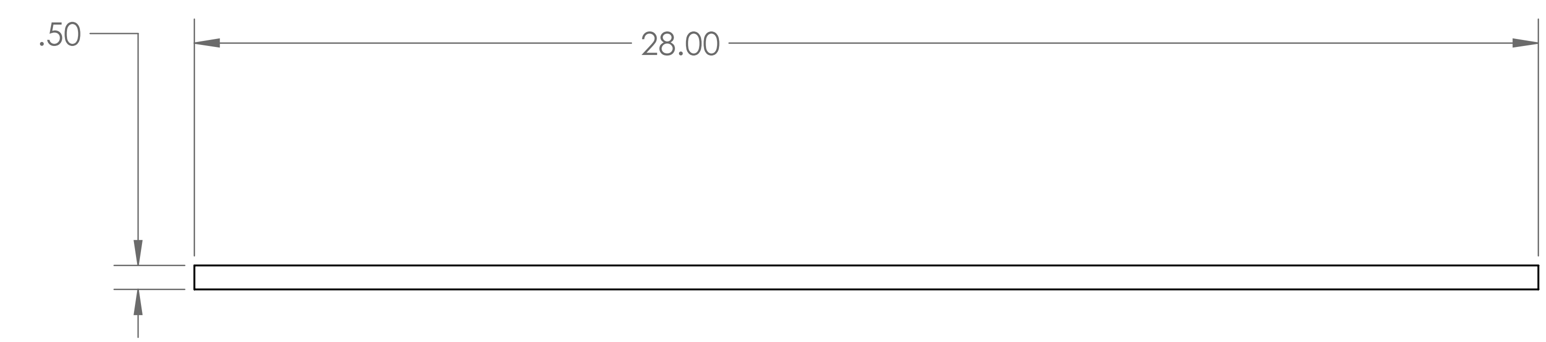
Fish Counting Device

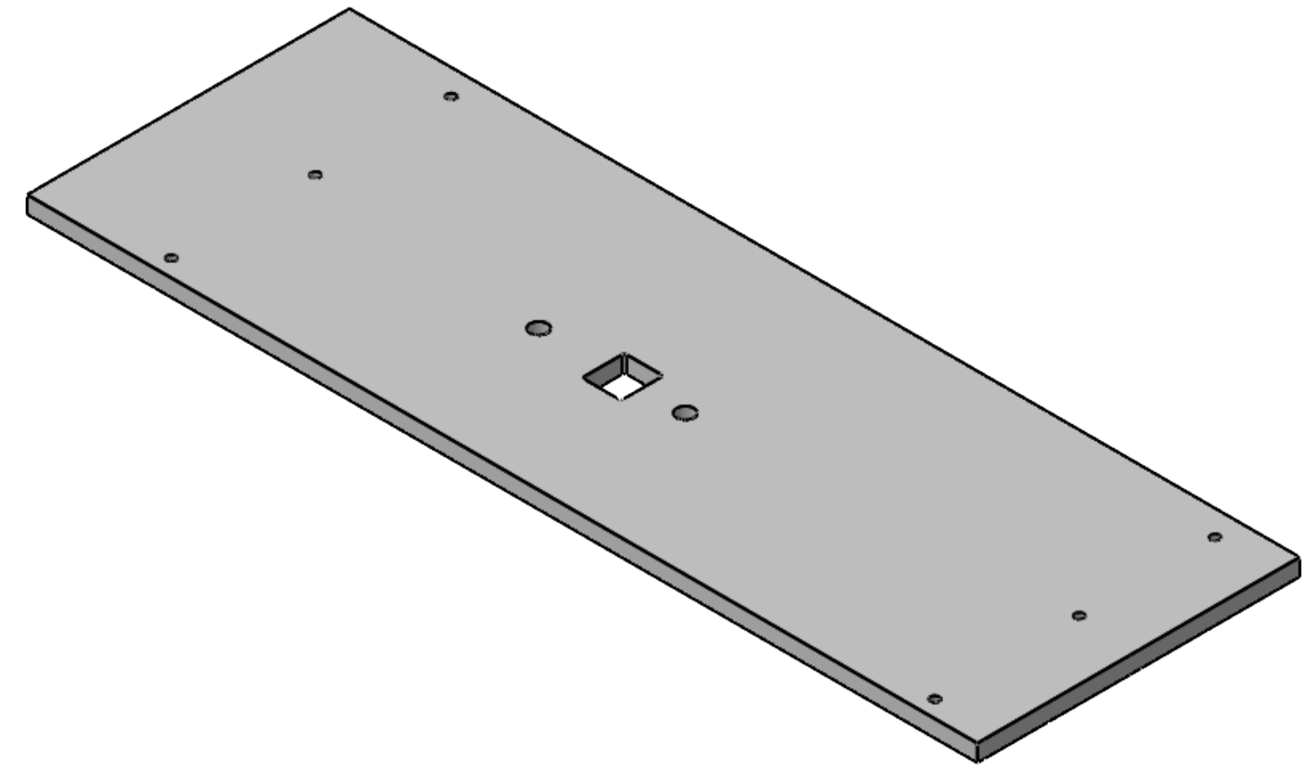
TITLE:

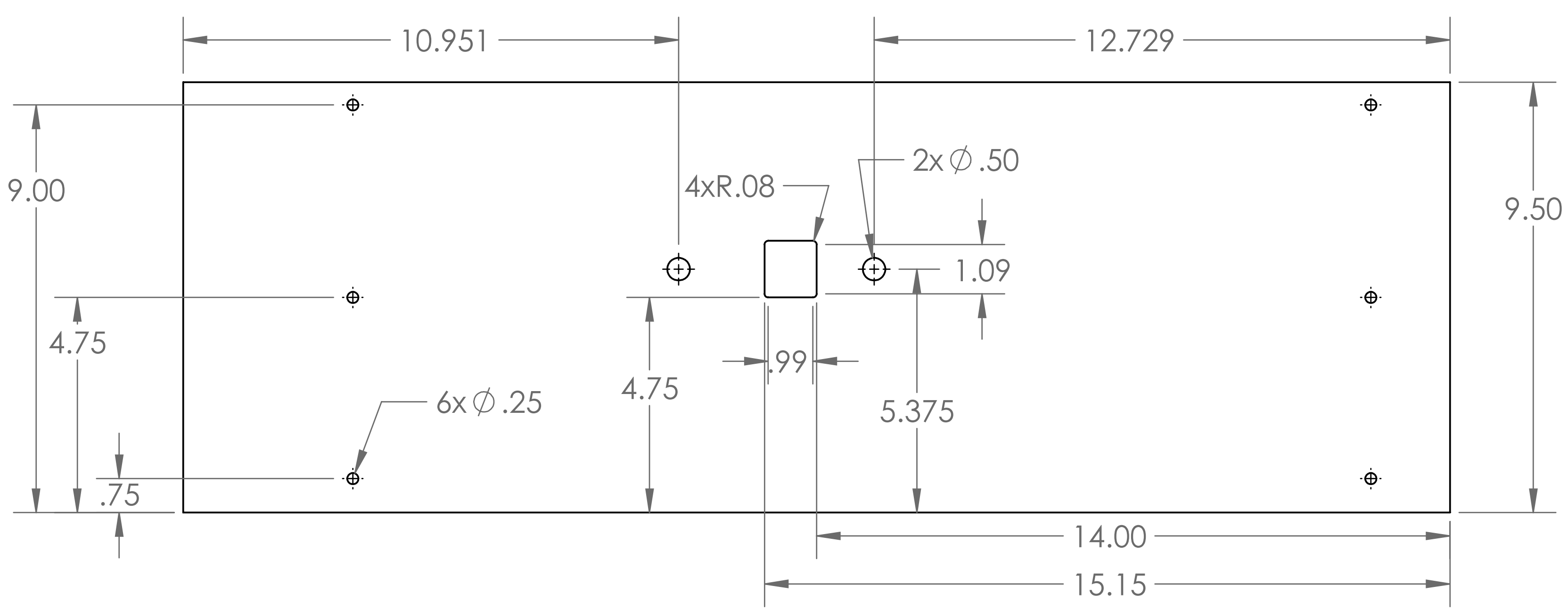
Roof 0.4 Section (TU) Part 2

SIZE	MATERIAL	REV
B	PVC Rigid	1

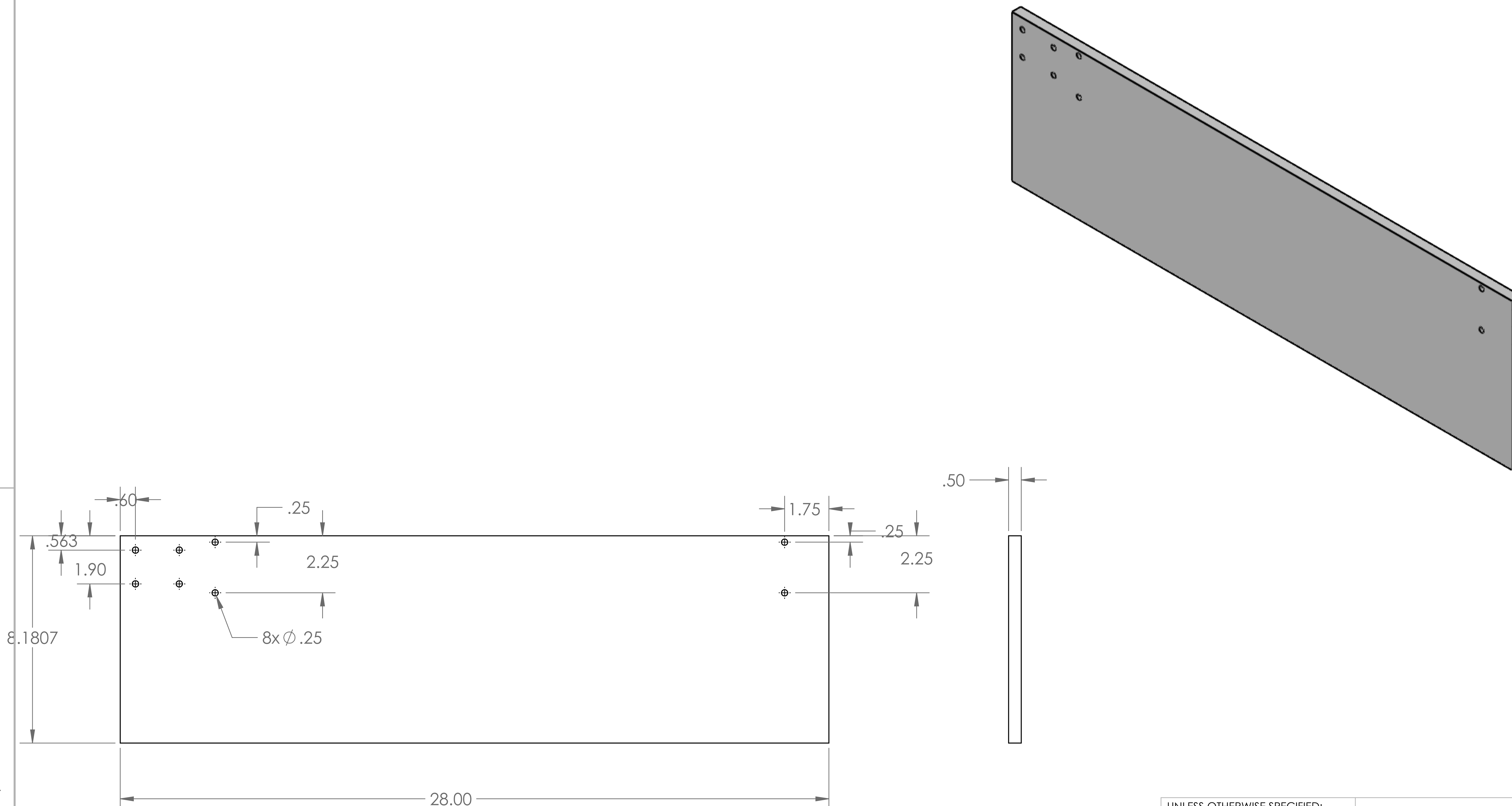
SCALE: 1:8 SHEET 1 OF 1



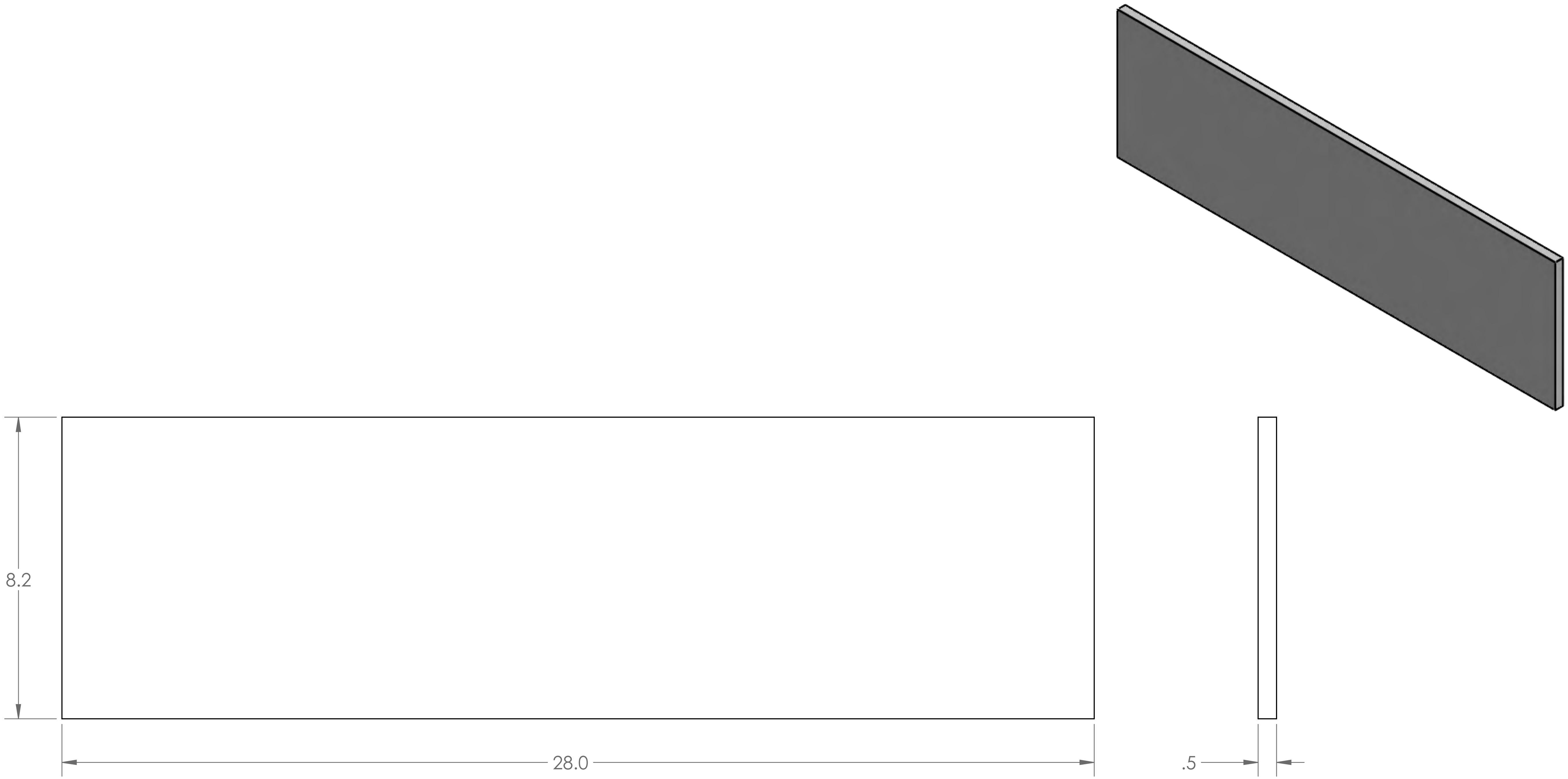




UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MM TOLERANCES: LINEAR X.X = ±0.5 X.XX = ±0.25 X.XXX = ±0.125 ANGULAR X = ±1° X.X = ±0.5°			**Fish Counting Device**		
	NAME	DATE	TITLE: Half Roof Horizontal		
DESIGNED	CLV	3/14/2026	SIZE	MATERIAL	REV
DRAWN	CLV	3/16/2026	**B**	PVC Rigid	1
MFG APPR.			SCALE: 1:8		SHEET 1 OF 1



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MM TOLERANCES: LINEAR X.X = ±0.5 X.XX = ±0.25 X.XXX = ±0.125 ANGULAR X = ±1° X.X = ±0.5°			**Fish Counting Device**	
			TITLE: Half Roof Vertical	
	NAME	DATE		
DESIGNED	CLV	3/13/2026	SIZE B	MATERIAL PVC Rigid	REV 1
DRAWN	CLV	3/16/2026		
MFG APPR.			SCALE: 1:8	SHEET 1 OF 1



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:

LINEAR	ANGULAR
X.X = ±0.1	X = ±1°
X.XX = ±0.03	X.X = ±0.5°
X.XXX = ±0.005	

	NAME	DATE
DESIGNED	CL	3/13/2026
DRAWN	NK	3/15/2026
MFG APPR.		

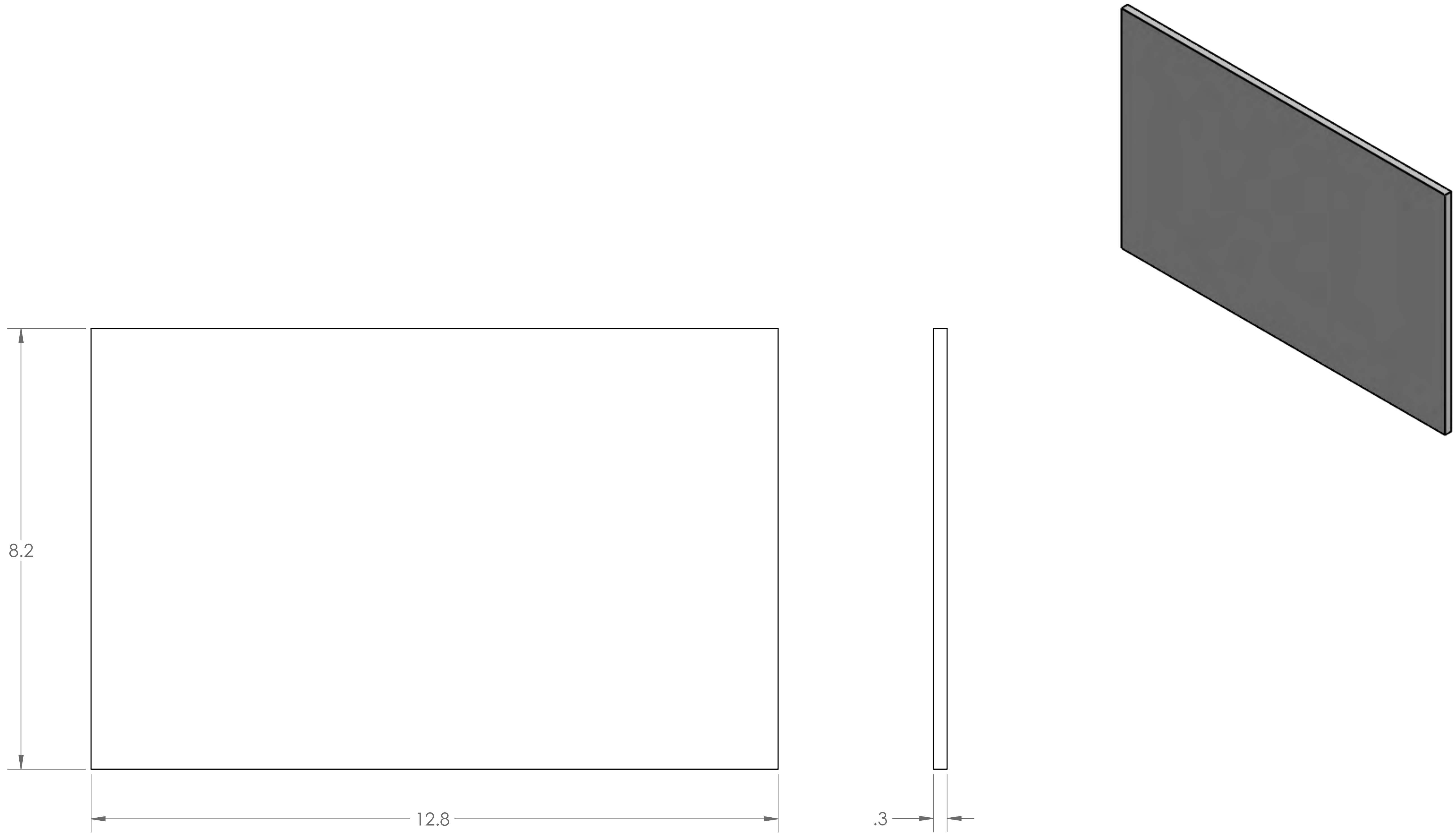
Fish Counting Device

TITLE:

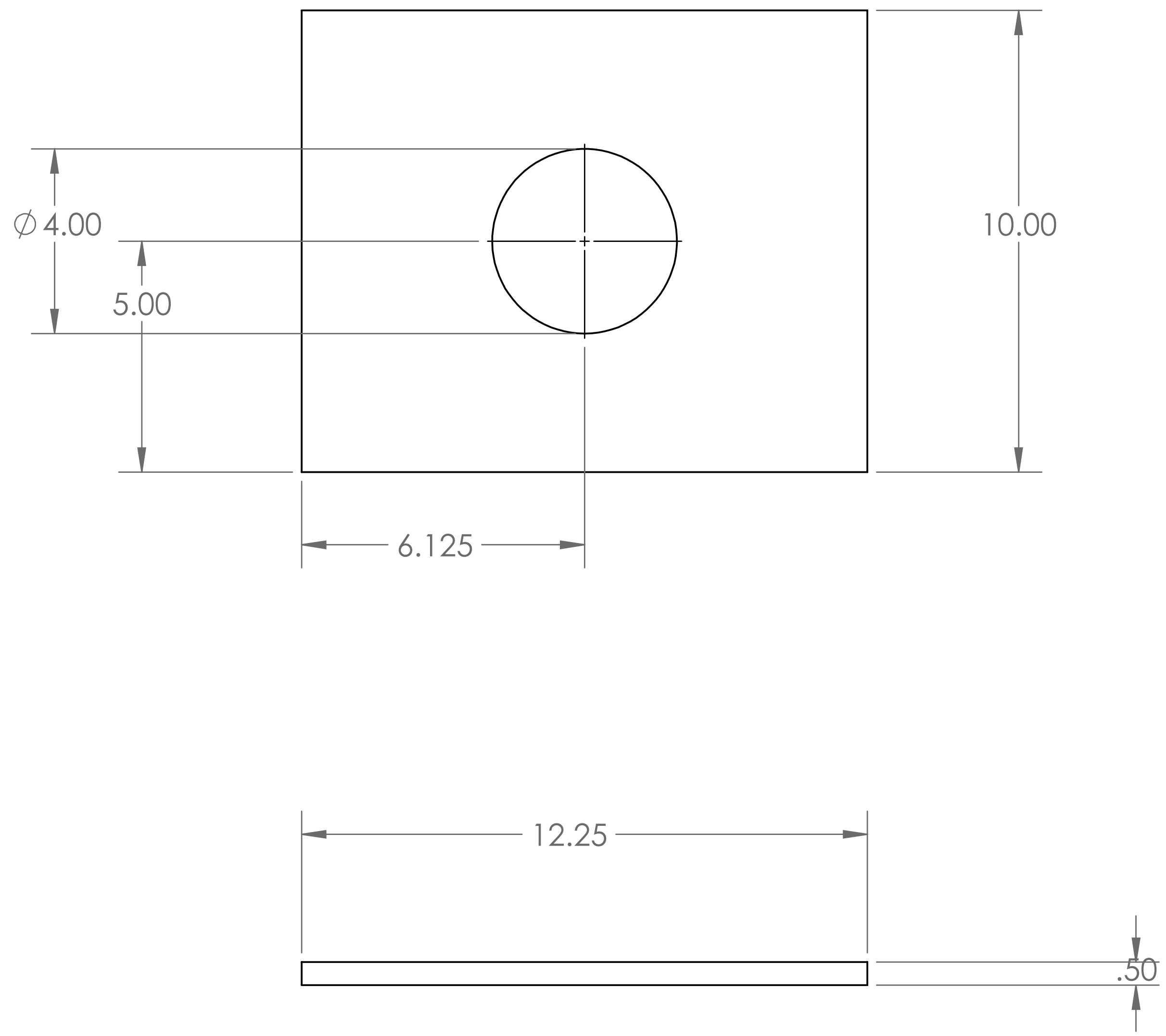
Side Wall on Lanes

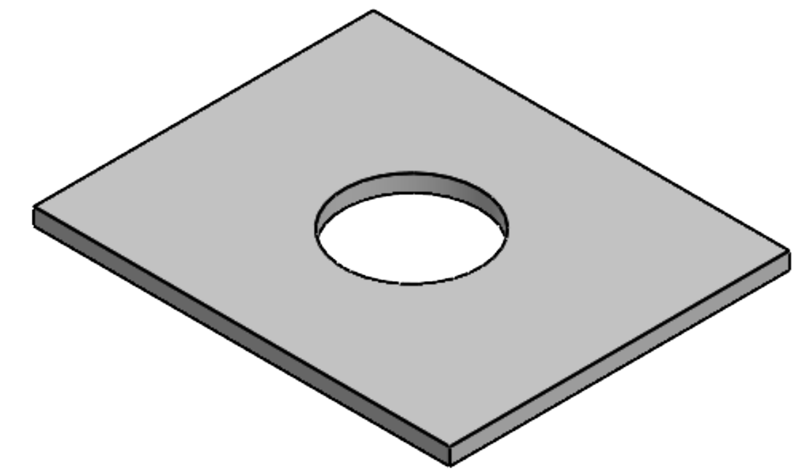
SIZE	MATERIAL	REV
B	PVC Rigid	1

SCALE: 1:3 SHEET 1 OF 1

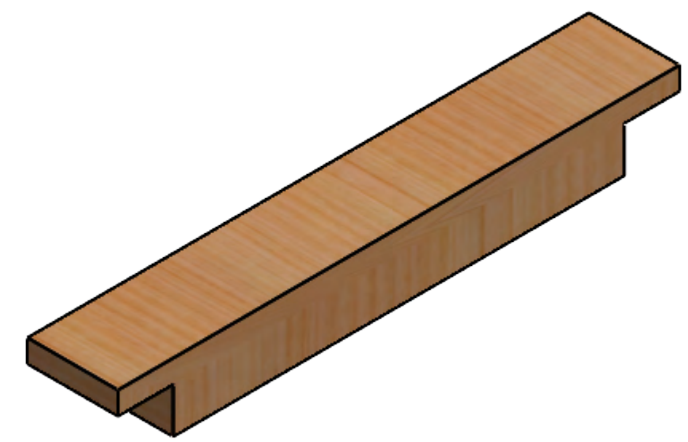


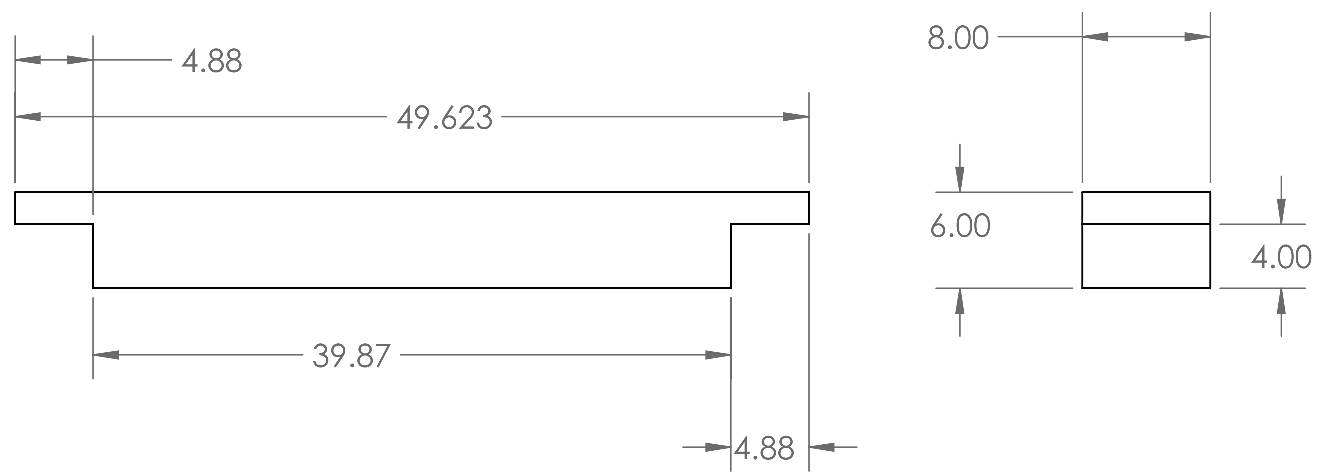
UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: LINEAR X.X = ±0.1 X.XX = ±0.03 X.XXX = ±0.005		ANGULAR X = ±1° X.X = ±0.5°	**Fish Counting Device**	
			TITLE: Back Wall	
	NAME	DATE		
DESIGNED	CL	3/13/2026	SIZE **B** MATERIAL PVC Rigid	REV 1
DRAWN	NK	3/15/2026		
MFG APPR.			SCALE: 1:2	SHEET 1 OF 1





UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MM TOLERANCES: LINEAR X.X = ±0.5 X.XX = ±0.25 X.XXX = ±0.125		ANGULAR X = ±1° X.X = ±0.5°	**Fish Counting Device**		
			TITLE: Bowl		
	NAME	DATE			
DESIGNED	CLV	3/15/2026	SIZE **B**	MATERIAL PVC Rigid	REV 1
DRAWN	CLV	3/16/2026			
MFG APPR.			SCALE: 1:4		SHEET 1 OF 1



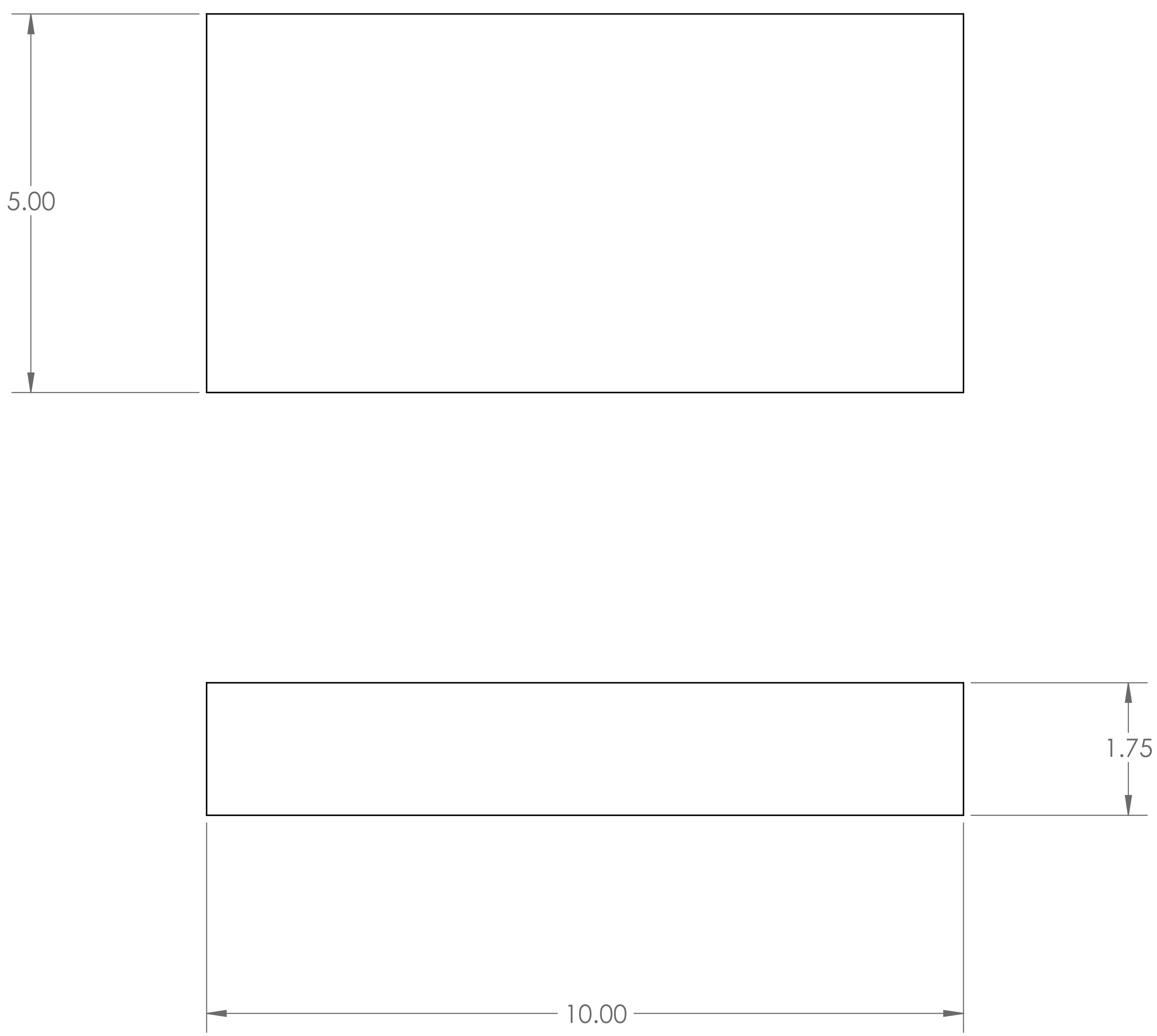


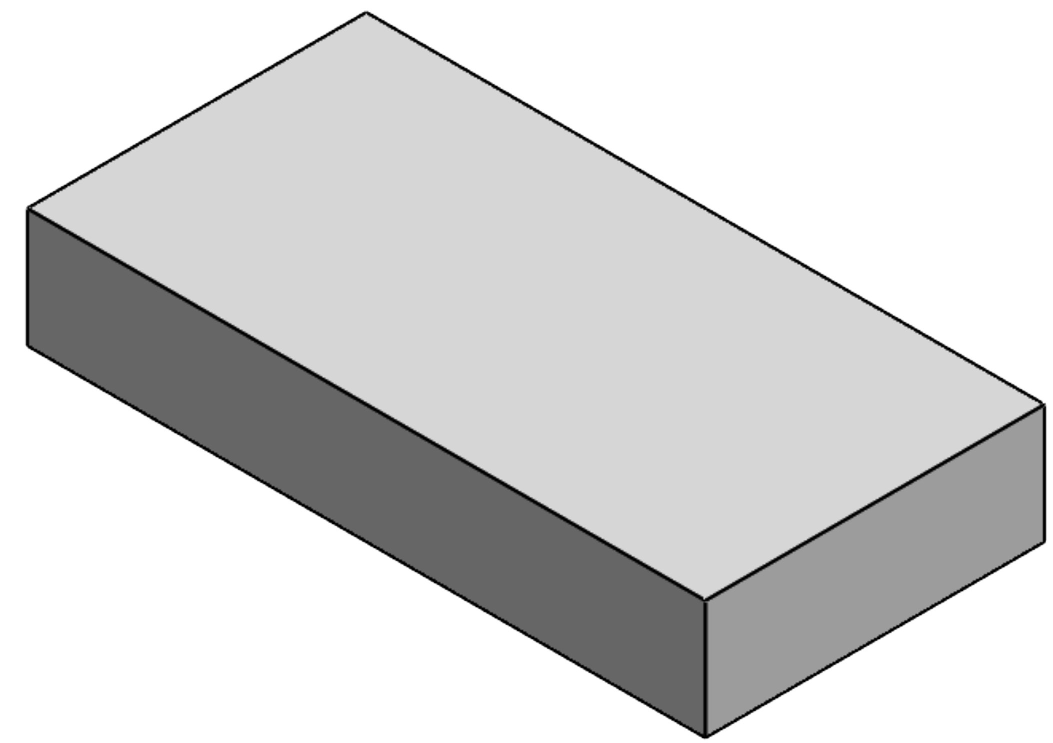
UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MM TOLERANCES: LINEAR X.X = ±0.5 X.XX = ±0.25 X.XXX = ±0.125	ANGULAR X = ±1° X.X = ±0.5°	**Fish Counting Device**
	NAME	DATE
DESIGNED	CLV	5/31/2026
DRAWN	CLV	6/5/2026
MFG APPR.		

TITLE:

Wood Block

SIZE	MATERIAL	REV
B	Cedar	1
	SCALE: 1:12	SHEET 1 OF 1





UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MM TOLERANCES: LINEAR: X.X = ±0.5, X.XX = ±0.25, X.XXX = ±0.125 ANGULAR: X = ±1°, X.X = ±0.5°			**Fish Counting Device**
	NAME	DATE	TITLE: Stabilizing Block
DESIGNED	CLV	3/14/2026	SIZE **B** · MATERIAL Wood · REV 1
DRAWN	CLV	3/16/2026	
MFG APPR.			SCALE: 1:2 · SHEET 1 OF 1